

Time Series Analysis

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Contents of Lecture

Spectral analysis methods:

- Discrete Fourier Transform
- Lomb-Scargle periodogram
- Least squares harmonic estimation
 - 1. Also applies when data are not equally spaced
 - 2. A linear trend is present
 - 3. Noise process of time series is considered (even non-stationary)
 - 4. Univariate/multivariate model
 - 5. Modulated harmonic estimation
 - 6. Multifrequency harmonic estimation
 - 7. Other applications



Lomb-Scargle Periodogram

- A physical variable $y = y(t_i), i = 1, \dots, m$ is measured at a set of times
- Signal is the sum of a signal and random error
$$y = y(t_i) = y_s(t_i) + e(t_i)$$
- Signal is contaminated with noise.
- Signal could be of periodic pattern and its multiple harmonics
- *Signal detection methods* resolve the problem.
 - Averaging with various periods
 - Least squares sine waves fitting
 - Periodogram analysis



Lomb-Scargle Periodogram

- DFT is

$$Y(\omega) = \sum_{j=1}^m y(t_j) e^{-i\omega t_j}$$

- Periodogram is conventionally defined (classical periodogram)

$$\begin{aligned} P(\omega) &= \frac{1}{m} |Y(\omega)|^2 \\ &= \frac{1}{m} \left| \sum_{j=1}^m y(t_j) e^{-i\omega t_j} \right|^2 \\ &= \frac{1}{m} \left[\left(\sum_{j=1}^m y(t_j) \cos \omega t_j \right)^2 + \left(\sum_{j=1}^m y(t_j) \sin \omega t_j \right)^2 \right] \end{aligned}$$

Why periodogram: if y contains a sinusoidal component of frequency ω_0 , then at and near $\omega = \omega_0$ the factors $y(t)$ and $e^{-i\omega t}$ are in phase and make a large contribution to the sums in the above equation.



Lomb-Scargle Periodogram

At other values ω the terms in the sum are **randomly positive and negative**, and the sum becomes small. Hence the presence of a sinusoid is indicated by a large value of P near one value of ω .

If observations are equally spaced at $\Delta t = 1, t_j = j, y_j = y(t_j)$, we have

$$P(\omega) = \frac{1}{m} \left| \sum_{j=1}^m y_j e^{-ij\omega} \right|^2$$

This expression is evaluated traditionally only at a special set of $\frac{m}{2}$ evenly spaced natural frequencies as

$$\omega_n = \frac{2\pi n}{T}, \quad \left\{ n = -\frac{m}{2}, \dots, \frac{m}{2} \right\}$$

$T = t_m = m$: total time span.



Lomb-Scargle Periodogram

The above power is evaluated at these frequencies:

- This can quickly be evaluated at $\omega_n \{ n = -\frac{m}{2}, \dots, \frac{m}{2} \}$ using FFT.
- Just enough information to recover original data
- All information is contained in positive frequencies $\{n = 0, 1, \dots, \frac{m}{2}\}$:
 $P(-\omega) = P(\omega)$
- Half of information has been thrown away by going from DFT to periodogram
- Evaluation of periodogram at intermediate frequencies is a kind of interpolation: adds no information

Therefore: $P(\omega_n)$ are independent random variables, $P(\omega)$ are dependent



Lomb-Scargle Periodogram

Critique of Periodogram:

- Statistical difficulties
- Spectral errors

If y is a normally distributed noise process, the relative variance

$$\frac{\sigma_{P(\omega)}}{P(\omega)}$$

is of order of unity.

No matter: how large m is, how dense samples are. Why?

Because the number of frequencies is also increased.



Lomb-Scargle Periodogram

If the process is

$$y(t_i) = A_0 \sin(\omega_0 t_i + \phi) + e(t_i), \quad i = 1, 2, \dots, m$$

Expected value of power due to signal (at ω_0) is

$$\bar{P} = E(P(\omega_0)) = \frac{mA_0^2}{4}$$

Expected value of power due to noise

$$\bar{P}_e = E(P_e) = \sigma_0^2$$

Signal-to-noise ratio (SNR)

$$SNR = \frac{\bar{P}}{\bar{P}_e} = \frac{mA_0^2}{4\sigma_0^2}$$

Increases proportionally to the number of samples m .



Lomb-Scargle Periodogram

Spectral errors:

Spectral leakage: Leakage to nearby frequencies (sidelobes) due to finite total interval over which the data are sampled

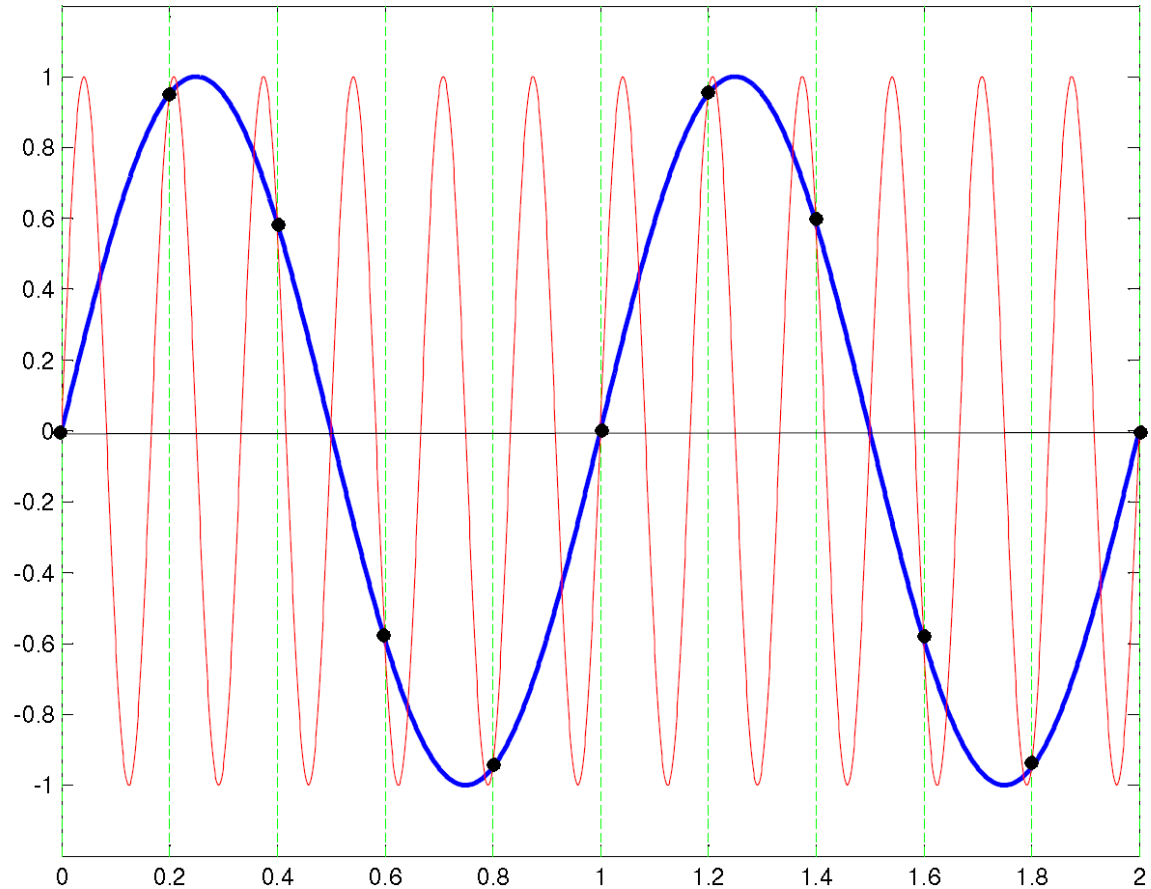
Spectral aliasing: Leakage to distant frequencies due to the finite size of the interval between samples.



Lomb-Scargle Periodogram and DFT

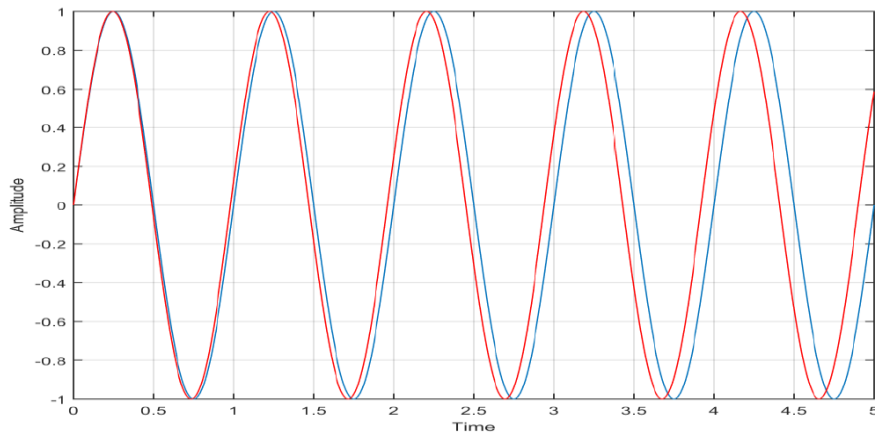
Spectral aliasing:

- Sampling interval: 0.2 (sec?)
- Sampling frequency: 5
- Period of correct signal: 1
- Frequency of correct signal: 1
- Period of aliased signal: $1/6$
- Frequency of aliased signal: 6
- Infinity number of aliased signal

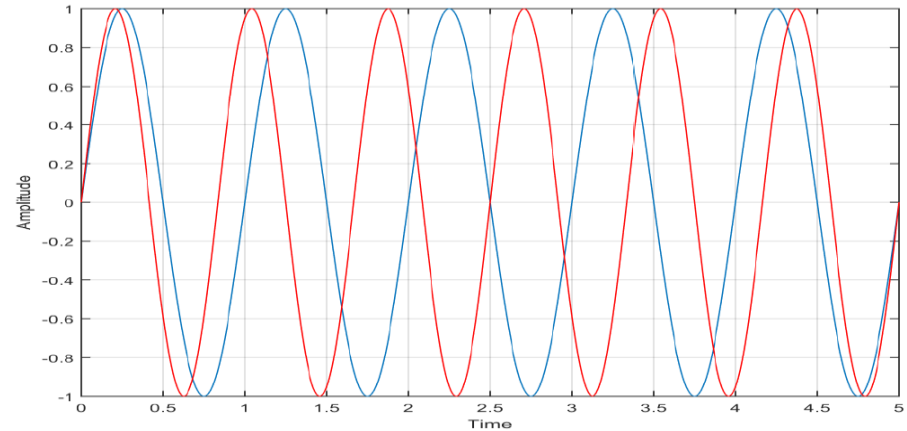


Lomb-Scargle Periodogram and DFT

Spectral leakage:



- Frequency of original signal: 1
- Frequency of other signal: 1.02



- Frequency of original signal: 1
- Frequency of other signal: 1.2



Lomb-Scargle Periodogram

The process is reformulated as

$$y(t_i) = a \cos(\omega_0 t_i) + b \sin(\omega_0 t_i) + e(t_i), \quad i = 1, 2, \dots, m$$

where σ_0^2 is the variance of the noise process. If the frequency ω is given, we may define the following notations

$$CC = \sum_{i=1}^m \cos^2(\omega t_i), \quad SS = \sum_{i=1}^m \sin^2(\omega t_i)$$

$$CS = \sum_{i=1}^m \cos(\omega t_i) \sin(\omega t_i)$$

$$YC = \sum_{i=1}^m y(t_i) \cos(\omega t_i), \quad YS = \sum_{i=1}^m y(t_i) \sin(\omega t_i)$$

Which leads to the normal equations

$$\begin{bmatrix} CC & CS \\ CS & SS \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} YC \\ YS \end{bmatrix}$$



Lomb-Scargle Periodogram

The reduction in the sum of squares is as follows:

$$P(\omega) = [YC \quad YS] \begin{bmatrix} CC & CS \\ CS & SS \end{bmatrix}^{-1} \begin{bmatrix} YC \\ YS \end{bmatrix}$$

The process may be reformulated as

$$y(t_i) = a \cos(\omega_0 t_i - \tau) + b \sin(\omega_0 t_i - \tau) + e(t_i), \quad i = 1, 2, \dots, m$$

and choosing τ such that CS .

This gives

$$P(\omega) = [YC \quad YS] \begin{bmatrix} CC & 0 \\ 0 & SS \end{bmatrix}^{-1} \begin{bmatrix} YC \\ YS \end{bmatrix}$$

or

$$P(\omega) = \frac{YC^2}{CC} + \frac{YS^2}{SS}$$



Lomb-Scargle Periodogram

or

$$P(\omega) = \frac{(\sum_{i=1}^m y(t_i) \cos(\omega t_i - \tau))^2}{\sum_{i=1}^m \cos^2(\omega t_i - \tau)} + \frac{(\sum_{i=1}^m y(t_i) \sin(\omega t_i - \tau))^2}{\sum_{i=1}^m \sin^2(\omega t_i - \tau)}$$

where

$$\tan(2\omega\tau) = \left(\sum_{i=1}^m \cos(2\omega t_i) \right) / \left(\sum_{i=1}^m \sin(2\omega t_i) \right)$$

The above formula can also be reformulated as

$$P(\omega) = \frac{(\sum_{i=1}^m y(t_i) \cos(\omega t_i))^2}{\sum_{i=1}^m \cos^2(\omega t_i)} + \frac{(\sum_{i=1}^m y(t_i) \sin(\omega t_i))^2}{\sum_{i=1}^m \sin^2(\omega t_i)}$$



Lomb-Scargle Periodogram

The power spectrum is of the form

$$P(\omega) = \left(\frac{1}{2}\right) \left(\frac{(\sum_{i=1}^m y(t_i) \cos(\omega t_i))^2}{\sum_{i=1}^m \cos^2(\omega t_i)} + \frac{(\sum_{i=1}^m y(t_i) \sin(\omega t_i))^2}{\sum_{i=1}^m \sin^2(\omega t_i)} \right)$$

For equally spaced data, we have $\sum_{i=1}^m \cos^2(\omega t_i) = \sum_{i=1}^m \sin^2(\omega t_i) = \frac{m}{2}$

$$P(\omega) = \left(\frac{1}{m}\right) \left[\left(\sum_{i=1}^m y(t_i) \cos(\omega t_i) \right)^2 + \left(\sum_{i=1}^m y(t_i) \sin(\omega t_i) \right)^2 \right]$$

We may define

$$C(\omega) = \frac{1}{\sqrt{m}} \sum_{i=1}^m y(t_i) \cos(\omega t_i), \quad S(\omega) = \frac{1}{\sqrt{m}} \sum_{i=1}^m y(t_i) \sin(\omega t_i)$$

To have $P(\omega) = C^2(\omega) + S^2(\omega)$

What is the distribution of $C(\omega)$ and $S(\omega)$?



Lomb-Scargle Periodogram

If $y(t_i) \sim N(0, \sigma_0^2)$, $i = 1, \dots, m$ are independent normal random variables, one has

$$\mathbf{E}(C(\omega)) = 0$$

and

$$\mathbf{D}(C(\omega)) = \mathbf{E}(C^2(\omega)) = \frac{1}{m} \mathbf{E} \left(\sum_{i=1}^m \sum_{j=1}^m y(t_i) y(t_j) \cos(\omega t_i) \cos(\omega t_j) \right)$$

or

$$\mathbf{D}(C(\omega)) = \frac{1}{m} \sum_{i=1}^m \sum_{j=1}^m \mathbf{E}(y(t_i) y(t_j) \cos(\omega t_i) \cos(\omega t_j))$$

or

$$\mathbf{D}(C(\omega)) = \frac{1}{m} \sum_{i=1}^m \mathbf{E}(y^2(t_i) \cos^2(\omega t_i))$$

or

$$\mathbf{D}(C(\omega)) = \frac{1}{m} \sum_{i=1}^m \mathbf{E}(y^2(t_i) \cos^2(\omega t_i))$$



Lomb-Scargle Periodogram

or

$$\mathbf{D}(C(\omega)) = \frac{1}{m} \sum_{i=1}^m \cos^2(\omega t_i) E(y^2(t_i))$$

or

$$\mathbf{D}(C(\omega)) = \frac{\sigma_0^2}{m} \sum_{i=1}^m \cos^2(\omega t_i)$$

or

$$\mathbf{D}(C(\omega)) = \frac{\sigma_0^2}{m} \times \frac{m}{2} = \frac{\sigma_0^2}{2}$$

A linear combination of normal random variables results in a normal variable. Therefore

$$C(\omega) \sim \mathbf{N}(0, \frac{\sigma_0^2}{2})$$

In a similar manner

$$S(\omega) \sim \mathbf{N}(0, \frac{\sigma_0^2}{2})$$



On distribution of random variables

If $Z = X^2 + Y^2$, where $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$ are two **independent** normal variables, then, Z is distributed as

$$Z \sim \frac{1}{2\sigma^2} e^{-\frac{z}{2\sigma^2}}$$

- **Exponential distribution with parameter $\lambda = \frac{1}{2\sigma^2}$**

If $Z = X^2 + Y^2$, where $X \sim N(0,1)$ and $Y \sim N(0,1)$ are two **independent** normal variables, then, is distributed as

$$Z \sim \chi^2(2) = \frac{1}{2} e^{-\frac{z}{2}}$$

- **Chi-squared distribution Chi-squared distribution with two degrees of freedom**
- **with two degrees of freedom**

If $y \sim N(\mu, Q_y)$, a multivariate normal distribution, then

$$Z = (y - \mu)^T Q_y^{-1} (y - \mu) \sim \chi^2(m)$$

- **Chi-squared distribution with m degrees of freedom**
- Department of Geomatics Engineering, University of Isfahan



Lomb-Scargle Periodogram

The power spectrum is then computed as

$$Z = P(\omega) = C^2(\omega) + S^2(\omega)$$

which has an exponential distribution as

$$Z = P(\omega) \sim \frac{1}{\sigma_0^2} e^{-\frac{Z}{\sigma_0^2}}$$

The variables $\frac{\sqrt{2}}{\sigma_0} C(\omega)$ and $\frac{\sqrt{2}}{\sigma_0} S(\omega)$ have standard normal distribution

$$\frac{\sqrt{2}}{\sigma_0} C(\omega) \sim N(0, 1) \quad \frac{\sqrt{2}}{\sigma_0} S(\omega) \sim N(0, 1)$$

which gives

$$\frac{2}{\sigma_0^2} P(\omega) = \frac{2}{\sigma_0^2} (C^2(\omega) + S^2(\omega)) \sim \chi^2(2)$$



Lomb-Scargle Periodogram

Natural frequencies and spectral window

If sampling is uneven, the best way is to use spectral response function, called also spectral window. The spectral window is used to describe the response of the entire data to a monochromatic (single frequency) sine wave.

The plain least-squares periodogram is then defined.

$$\min_{\substack{A \geq 0 \\ \phi \in [0, 2\pi]}} f(A, \phi) = \sum_{j=1}^m (y(t_j) - A_0 \sin(\omega t_j + \phi))^2$$

We may define

$$y = \begin{bmatrix} y(t_1) \\ \vdots \\ y(t_m) \end{bmatrix}, \quad A = \begin{bmatrix} \cos \omega t_1 & \sin \omega t_1 \\ \vdots & \vdots \\ \cos \omega t_m & \sin \omega t_m \end{bmatrix}, \quad x = \begin{bmatrix} A_0 \sin \phi = a \\ A_0 \cos \phi = b \end{bmatrix}$$



Lomb-Scargle Periodogram

We obtain

$$\min_x f(x) = \sum_{j=1}^m \|y - Ax\|^2$$

which gives

$$\hat{x} = (A^T A)^{-1} A^T y$$

Plain least squares periodogram (Least squares power spectrum) is defined

$$P_{LS}(\omega) = (A\hat{x})^T (A\hat{x}) = \hat{x}^T A^T A \hat{x}$$

or

$$P_{LS}(\omega) = y^T A (A^T A)^{-1} A^T A (A^T A)^{-1} A^T y = y^T A (A^T A)^{-1} A^T y = u^T N^{-1} u$$

where

$$N = A^T A, \quad u = A^T y$$



Lomb-Scargle Periodogram

If $y \sim \mathbf{N}(0, I)$ it follows

$$u \sim \mathbf{N}(0, A^T A) = \mathbf{N}(0, N)$$

$$P_{LS}(\omega) = u^t N^{-1} u \sim \chi^2(2)$$

If $y \sim \mathbf{N}(0, \sigma^2 I)$ it follows

$$u \sim \mathbf{N}(0, \sigma^2 A^T A) = \mathbf{N}(0, \sigma^2 N)$$

$$\sigma^{-2} P_{LS}(\omega) = \sigma^{-2} u^t N^{-1} u \sim \chi^2(2)$$



Lomb-Scargle Periodogram

Invariance property to time translation

If we reformulate

$$\min_{\substack{A \geq 0 \\ \phi \in [0, 2\pi]}} f(A, \phi) = \sum_{j=1}^m (y(t_j) - A_0 \sin(\omega(t_j + \tau) + \phi))^2$$

We may define

$$y = \begin{bmatrix} y(t_1) \\ \vdots \\ y(t_m) \end{bmatrix}, \quad A = \begin{bmatrix} \cos \omega t_1 & \sin \omega t_1 \\ \vdots & \vdots \\ \cos \omega t_m & \sin \omega t_m \end{bmatrix}, \quad x' = \begin{bmatrix} A_0 \sin(\phi + \omega \tau) = a' \\ A_0 \cos(\phi + \omega \tau) = b' \end{bmatrix}$$

Plain least squares periodogram (Least squares power spectrum) is defined

$$P_{LS}(\omega) = (A\hat{x}')^T (A\hat{x}') = \hat{x}'^T A^T A \hat{x}'$$

or

$$P_{LS}(\omega) = y^T A (A^T A)^{-1} A^T y$$



Lomb-Scargle Periodogram

Summary

- The Lomb-Scargle Periodogram and the Least-Squares Periodogram give the same estimate (because the best least-squares fitting is unique).
- PLS is simpler than $P_{(\text{Lomb-Scargle})}$ but with the same nice properties, we recommend using PLS
- Has a clear statistical description using statistical testing
- Has the invariance property to time translation
- Further development of PLS is possible



Least squares harmonic estimation

Hypothesis testing in a linear model:

The m -vector of observables $\underline{y}$ is assumed to be normally distributed

$$\underline{y} \sim N_m(E(\underline{y}), Q_{yy})$$

The following two composite hypotheses on the expectation of $\underline{y}$ are put forward

$$H_o : E(\underline{y}) = Ax$$

$$H_a : E(\underline{y}) = Ax + C_y \nabla$$

Either accepts or rejects null hypothesis



Least squares harmonic estimation

Null hypothesis is rejected if:

$$\text{reject } H_o \text{ if } \hat{e}_o^T Q_{yy}^{-1} \hat{e}_o - \hat{e}_a^T Q_{yy}^{-1} \hat{e}_a > k_\alpha$$

Four representations for left hand side:

$$1) \quad \underline{T}_q = \underline{\hat{e}}_o^T Q_{yy}^{-1} \underline{\hat{e}}_o - \underline{\hat{e}}_a^T Q_{yy}^{-1} \underline{\hat{e}}_a$$

$$2) \quad T_q = (\hat{y}_o - \hat{y}_a)^T Q_{yy}^{-1} (\hat{y}_o - \hat{y}_a)$$

$$3) \quad T_q = \hat{\nabla}^T C_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} C_y \hat{\nabla}$$

$$4) \quad T_q = \hat{e}_o^T Q_{yy}^{-1} C_y (C_y^T Q_{yy}^{-1} Q_{\hat{e}_o \hat{e}_o} Q_{yy}^{-1} C_y)^{-1} C_y^T Q_{yy}^{-1} \hat{e}_o$$



Least squares harmonic estimation

If the original observations are normally distributed, one has

$$H_o : \underline{T}_q \sim \chi^2(q, 0) \text{ and } H_a : \underline{T}_q \sim \chi^2(q, \lambda)$$

with noncentrality parameter

$$\lambda = \nabla^T Q_{\hat{\nabla}\hat{\nabla}}^{-1} \nabla = \nabla^T C_y^T Q_{yy}^{-1} Q_{\hat{e}_o\hat{e}_o} Q_{yy}^{-1} C_y \nabla$$

- This can have a variety of applications in functional model identification.
 - Outliers
 - Offsets
 - Periodic patterns



Least squares harmonic estimation

Identification of deterministic effects in GNSS time series:

$$H_0: E(y) = Ax, D(y) = Qy \quad \text{vs.} \quad H_a: E(y) = Ax + A_k x_k, D(y) = Qy$$

Example:

For a linear regression model with a periodic function we have

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}$$

$$A_k = \begin{bmatrix} \cos \omega_k t_1 & \sin \omega_k t_1 \\ \cos \omega_k t_2 & \sin \omega_k t_2 \\ \vdots & \vdots \\ \cos \omega_k t_m & \sin \omega_k t_m \end{bmatrix}$$



Least squares harmonic estimation

The test statistic (least squares power spectrum) is

$$T_2 = P(\omega_k) = \hat{e}^T Q_y^{-1} A_k (A_k^T Q_y^{-1} P_A^\perp A_k) A_k^T Q_y^{-1} \hat{e}$$

where

$$P_A^\perp = I - A(A^T Q_y^{-1} A)^{-1} A^T Q_y^{-1}$$

is an orthogonal projector, and

$$\hat{e} = P_A^\perp y$$

is the residual vector under the null hypothesis.



Least squares harmonic estimation

Hypothesis Testing

Under null hypothesis

$$T_2 \sim \chi^2(2, 0)$$

versus alternative hypothesis

$$T_2 \sim \chi^2(2, \lambda)$$

Null hypothesis is rejected if

$$T_2 > \chi^2_{2, \alpha}$$

Presence of a periodic pattern is then confirmed in the data.



Least squares harmonic estimation

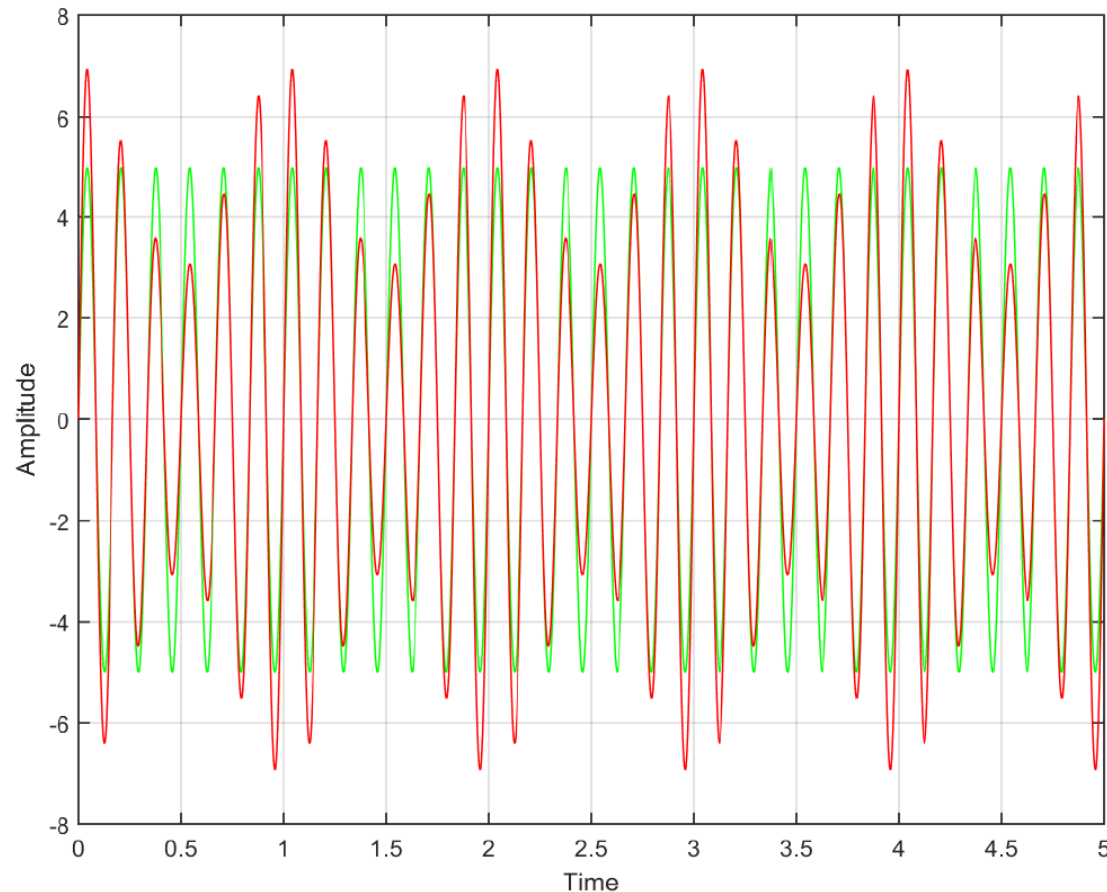
Modulated LS-HE

In its basic form, amplitude modulation consists of the signal

$$y(t) = A(t) \sin(\omega t + \phi)$$

in which the amplitude $A(t)$ of the signal is a function of time. When $A(t)$ is also a sinusoidal wave, amplitude modulation produces a signal with power concentrated at the carrier frequency and in two adjacent sidebands, which is called double-sideband amplitude modulation.

$$5 \sin(6t) \text{ vs. } (5 + 2 \sin(t))\sin(6t)$$



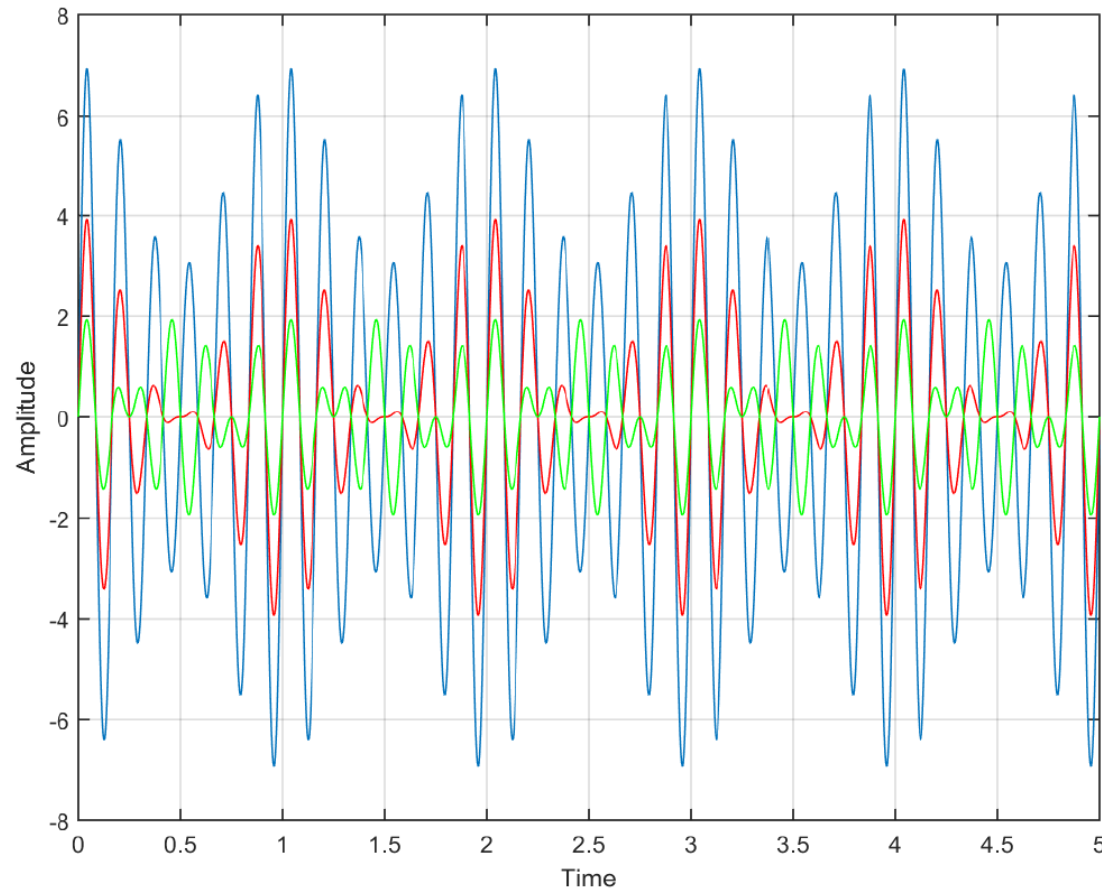
Least squares harmonic estimation

Modulated LS-HE

This can, for instance, be due to the diurnal signal with frequency ω_2 modulated with the annual signal with frequency ω_1 , i.e., the amplitude of the diurnal signal changes periodically over one year.

The modulated signal can have three components:

- The original wave
- Two sinusoidal waves whose frequencies are slightly above and below the original frequency.



Least squares harmonic estimation

Modulated LS-HE

To understand how the harmonic estimation method works for such modulated signals, consider that the amplitude modulation is created by forming the following product:

$$y(t) = (A + M \sin(\omega_1 t + \phi_1)) \sin(\omega_2 t + \phi_2)$$

where

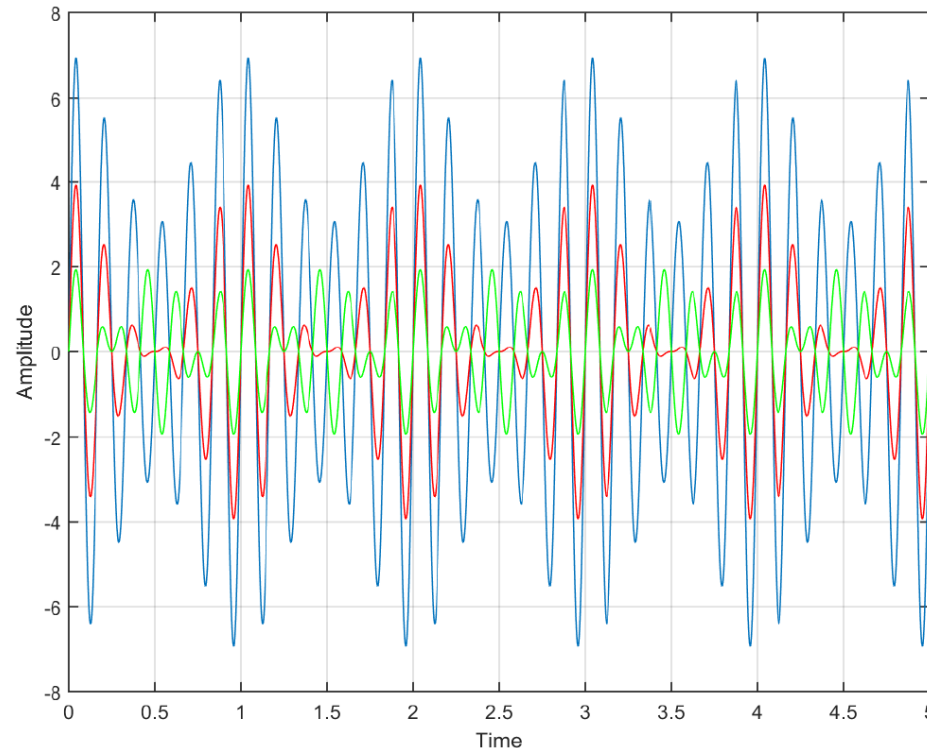
$$y'(t) = A \sin(\omega_2 t + \phi_2)$$

is the original wave, and

$$y'' = M \sin(\omega_1 t + \phi_1) \sin(\omega_2 t + \phi_2)$$

is the modulated wave.

How to formulate it in LS-HE?



Least squares harmonic estimation

Modulated LS-HE

We may use the identities:

$$\sin \alpha \sin \beta = \frac{1}{2} [\cos(\alpha - \beta) - \cos(\alpha + \beta)]$$

and

$$\cos \alpha = \sin\left(\frac{\pi}{2} - \alpha\right)$$

which give

$$\sin \alpha \sin \beta = \frac{1}{2} \left[\sin\left(\frac{\pi}{2} - \alpha + \beta\right) - \sin\left(\frac{\pi}{2} - \alpha - \beta\right) \right]$$

or

$$\sin \alpha \sin \beta = \frac{1}{2} \left[\sin\left(\beta - \alpha + \frac{\pi}{2}\right) + \sin\left(\alpha + \beta - \frac{\pi}{2}\right) \right]$$

For modulated signal we had:

$$y'' = M \sin(\omega_1 t + \phi_1) \sin(\omega_2 t + \phi_2)$$



Least squares harmonic estimation

Modulated LS-HE

This will then provides

$$y'' = \frac{M}{2} \left[\sin((\omega_2 - \omega_1)t + \phi_2 - \phi_1 + \frac{\pi}{2}) + \sin((\omega_2 + \omega_1)t + \phi_2 + \phi_1 - \frac{\pi}{2}) \right]$$

or

$$y'' = \frac{M}{2} \left[\sin(\omega_d t + \phi_d + \frac{\pi}{2}) + \sin(\omega_s t + \phi_s - \frac{\pi}{2}) \right]$$

or

$$y'' = y_1 + y_2 = \frac{M}{2} [\sin(\omega_s t + \phi_s) + \sin(\omega_d t + \phi_d)]$$

where

$$y_1 = \frac{M}{2} \sin(\omega_s t + \phi_s) = a_s \cos \omega_s t + b_s \sin \omega_s t$$

and

$$y_2 = \frac{M}{2} \sin(\omega_d t + \phi_d) = a_d \cos \omega_d t + b_d \sin \omega_d t$$

which gives

$$y'' = y_1 + y_2 = a_s \cos \omega_s t + b_s \sin \omega_s t + a_d \cos \omega_d t + b_d \sin \omega_d t$$



Least squares harmonic estimation

Modulated LS-HE

Identification of deterministic effects in GNSS time series:

$$H_0: E(y) = Ax, D(y) = Qy \quad \text{vs.} \quad H_a: E(y) = Ax + A_k x_k, D(y) = Qy$$

where

$$A_k = \begin{bmatrix} \cos \omega_s t_1 & \sin \omega_s t_1 & \cos \omega_d t_1 & \sin \omega_d t_1 \\ \cos \omega_s t_2 & \sin \omega_s t_2 & \cos \omega_d t_2 & \sin \omega_d t_2 \\ \vdots & \vdots & \vdots & \vdots \\ \cos \omega_s t_m & \sin \omega_s t_m & \cos \omega_d t_m & \sin \omega_d t_m \end{bmatrix}, \quad x_k = \begin{bmatrix} a_s \\ b_s \\ a_d \\ b_d \end{bmatrix},$$

The test statistic (least squares modulated power spectrum) is

$$T_4 = P(\omega_k) = \hat{e}^T Q_y^{-1} A_k (A_k^T Q_y^{-1} P_A^\perp A_k) A_k^T Q_y^{-1} \hat{e}$$



Least squares harmonic estimation

Modulated LS-HE: Hypothesis Testing

Under null hypothesis

$$T_4 \sim \chi^2(4, 0)$$

versus alternative hypothesis

$$T_4 \sim \chi^2(4, \lambda)$$

Null hypothesis is rejected if

$$T_4 > \chi_{4,\alpha}^2$$

Presence of a modulated periodic pattern is then confirmed in the data.



Least squares harmonic estimation

Multiharmonic LS-HE

Another feature of the LS-HE is the use of multiharmonic model. We aim to detect a periodic pattern using all its constitutive harmonics. From the Fourier series decomposition of a periodic function, it is well known that any periodic pattern is composed of a signal with a principal frequency along with its higher harmonics. We therefore may consider to detect the principal frequency of the periodic pattern using all of its harmonics simultaneously. The method can be applied to a single time series in univariate analysis or to multiple time series in multivariate analysis. Let us assume that a periodic pattern with the principal frequency of ω and its higher harmonics $2\omega, \dots, k\omega$ is present in a univariate time series. The total signal due to this periodic pattern is composed of p sinusoidal waves as

$$y(t) = \sum_{i=1}^k a_i \cos i\omega t + b_i \sin i\omega t$$

where $i\omega$ is the frequency of the i th harmonic.



Least squares harmonic estimation

Multiharmonic LS-HE: Example

$K=3$

=====

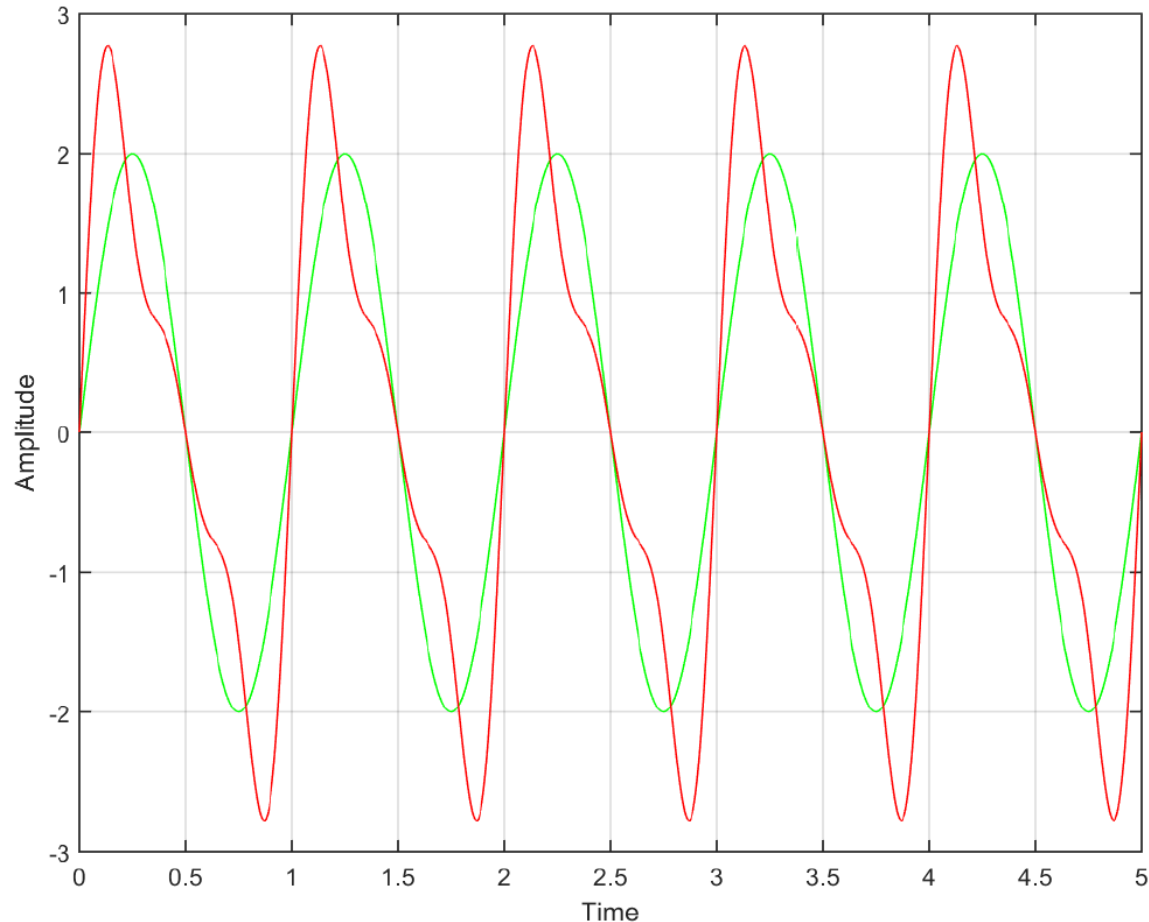
$A_1 = 2$

$A_2 = 1$

$A_3 = 0.5$

Why this is useful?

How they can be incorporated
in a single model?



Least squares harmonic estimation

Multiharmonic LS-HE

Identification of deterministic effects in GNSS time series:

$$H_0: E(y) = Ax, D(y) = Qy \quad \text{vs.} \quad H_a: E(y) = Ax + A_k x_k, D(y) = Qy$$

where

$$A_k = \begin{bmatrix} \cos(\omega t_1) & \sin(\omega t_1) & \cos(2\omega t_1) & \sin(2\omega t_1) & \cdots & \cos(k\omega t_1) & \sin(k\omega t_1) \\ \cos(\omega t_2) & \sin(\omega t_2) & \cos(2\omega t_2) & \sin(2\omega t_2) & \cdots & \cos(k\omega t_2) & \sin(k\omega t_2) \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \cos(\omega t_m) & \sin(\omega t_m) & \cos(2\omega t_m) & \sin(2\omega t_m) & \cdots & \cos(k\omega t_m) & \sin(k\omega t_m) \end{bmatrix}$$

The test statistic (least squares modulated power spectrum) is

$$T_{2k} = P(\omega_k) = \hat{e}^T Q_y^{-1} A_k (A_k^T Q_y^{-1} P_A^\perp A_k) A_k^T Q_y^{-1} \hat{e} \sim \chi^2(2k, 0)$$



Least squares harmonic estimation

Multivariate LS-HE

If in the linear model, instead of one time series y , there exist several (r) time series with identical A and Qy , and the corresponding parameter vectors have to be determined, the model is referred to as a multivariate linear model.

Examples include:

- We may think of the TEC time series at different latitudes.
- We may think of multiple GPS position time series
- We may think of multiple sea level height time series

When dealing with multiple time series, one may apply the LS-HE to obtain the **multivariate power spectrum** for detecting the common-mode periodic patterns of all series.



Least squares harmonic estimation

Multivariate LS-HE

For a multivariate model, model of observation equations is generalized to

Null hypothesis:

$$\mathbf{E}(\text{vec}(Y)) = (I \otimes A)\text{vec}(X), \quad \mathbf{D}(\text{vec}(Y)) = \Sigma \otimes Q$$

Alternative hypothesis:

$$\mathbf{E}(\text{vec}(Y)) = (I \otimes A)\text{vec}(X) + (I \otimes A_k)\text{vec}(X_k), \quad \mathbf{D}(\text{vec}(Y)) = \Sigma \otimes Q$$

where

$$Y = [y_1 \ y_2 \ \dots \ y_r]$$

$$X = [x_1 \ x_2 \ \dots \ x_r]$$

$$X_k = [x_{1k} \ x_{2k} \ \dots \ x_{rk}]$$



Least squares harmonic estimation

Multivariate LS-HE

We have to substitute the terms in the univariate model as follows:

$$\left\{ \begin{array}{l} I \otimes A \rightarrow A \\ I \otimes A_k \rightarrow A_k \\ \Sigma \otimes Q \rightarrow Q_y \\ \text{vec}(Y) \rightarrow y \\ \text{vec}(\hat{E}) \rightarrow \hat{e} \end{array} \right. \quad T_2 = P(\omega_k) = \hat{e}^T Q_y^{-1} A_k (A_k^T Q_y^{-1} P_A^\perp A_k) A_k^T Q_y^{-1} \hat{e}$$

The test statistics becomes

$$T_{2r} = \text{tr} \left(\hat{E}^T Q^{-1} A_k (A_k^T Q^{-1} P_A^\perp A_k)^{-1} A_k^T Q^{-1} \hat{E} \hat{\Sigma}^{-1} \right) \sim \chi^2(2r)$$

where

$$\hat{\Sigma} = \frac{\hat{E}^T Q^{-1} \hat{E}}{m - n}$$



Least squares harmonic estimation

Multivariate LS-HE

Multivariate least squares harmonic can handle the three following cases:

- A pure harmonic model

$$T_{2r} \sim \chi^2(2r)$$

- A modulated harmonic model

$$T_{4r} \sim \chi^2(4r)$$

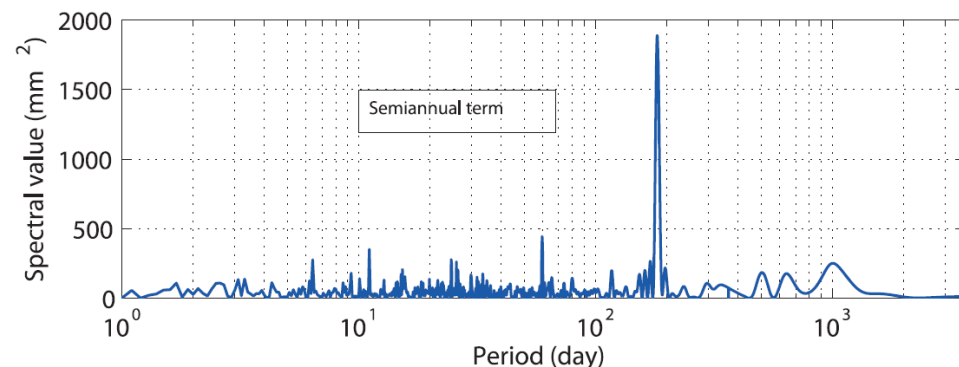
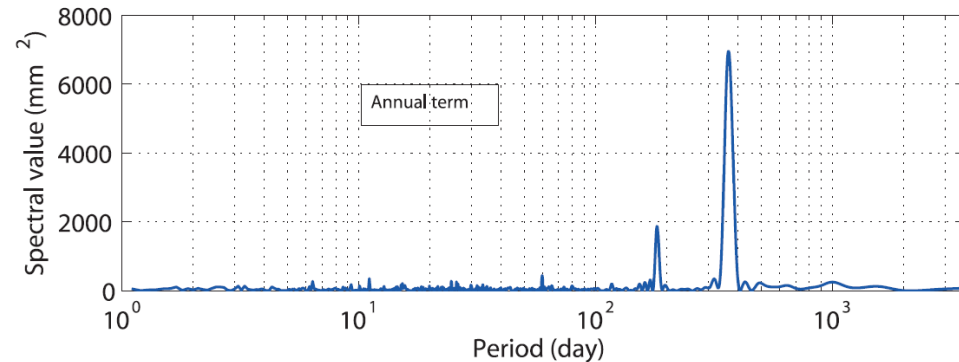
- A multiharmonic model

$$T_{kr} \sim \chi^2(kr)$$



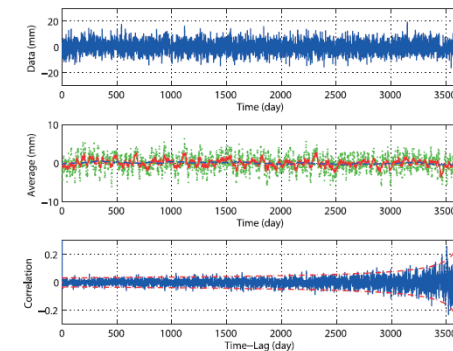
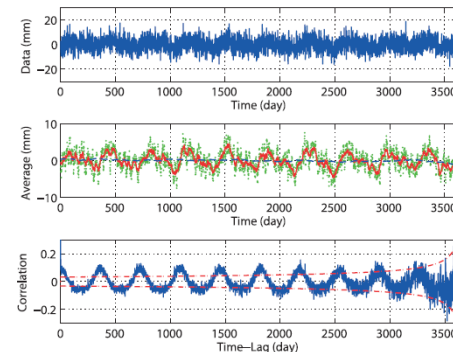
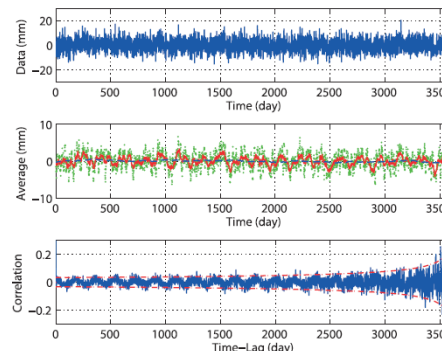
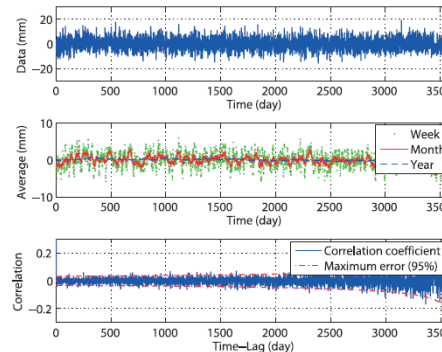
LS-HE: Simulated data

- We have used simulated data. We have simulated a 10-year time-series (daily samples) containing only white noise with a standard deviation of 5 mm.
- Two sinusoidal functions with amplitude of 2 and 1 mm, respectively, for the annual and semiannual term, have been added to the data; a period of 365.25 days, to pretend the annual term, and 182.625 days for semiannual term.
- The goal now is to use the harmonic estimation method to identify the frequencies (or periods) of these harmonic functions. The process has been repeated 100 times and it follows with the aforementioned assumptions that the (empirical) standard deviation of the periods of these harmonic functions is 1.4 days for annual term and 0.7 day for semiannual term.
- It is well known that increasing the amplitude of harmonic functions and increasing the time-series length gives more precise results for the period (or frequency) of the sinusoidal functions.
- Figure shows one typical example of application of LS-HE to find the periods of harmonic functions. In the first step (top figure), the annual term (period of 364.7 days) is detected and in the second step the semiannual signal) period of 182.4 days.



LS-HE: Simulated data

- A correlogram portrays the autocorrelation versus time-lag. The coefficient at lag zero equals one by definition. If a white noise process can describe the stochastic characteristic of the series, then all other coefficients should be approximately zero because a sample at one epoch is not correlated with one at any other epoch.
- Figure shows the typical example of the simulated data corresponding to the previous figure. In each graph, the top figure is the time series itself, the middle shows the running average of the series over one week, one month and one year, and the bottom gives the autocorrelation coefficients obtained from LS-VCE.
- In case of white noise, autocorrelation function behaves randomly around zero. When, in addition, both the annual and semiannual terms are included, the auto-correlation function shows a periodic behavior which resembles the periodic behavior of the annual signal. Note that the amplitude of the annual signal is twice of that of the semiannual term.
- Therefore if there are unmodelled effects in data, they can mistakenly be interpreted as time correlation. When we detect the annual term and include the signal in the functional model, the autocorrelation function shows still a periodic behavior. This behavior is due to the presence of the semiannual term in the data.
- After removing this signal, the autocorrelation function obtained is very similar to what we saw for the case of pure white noise.

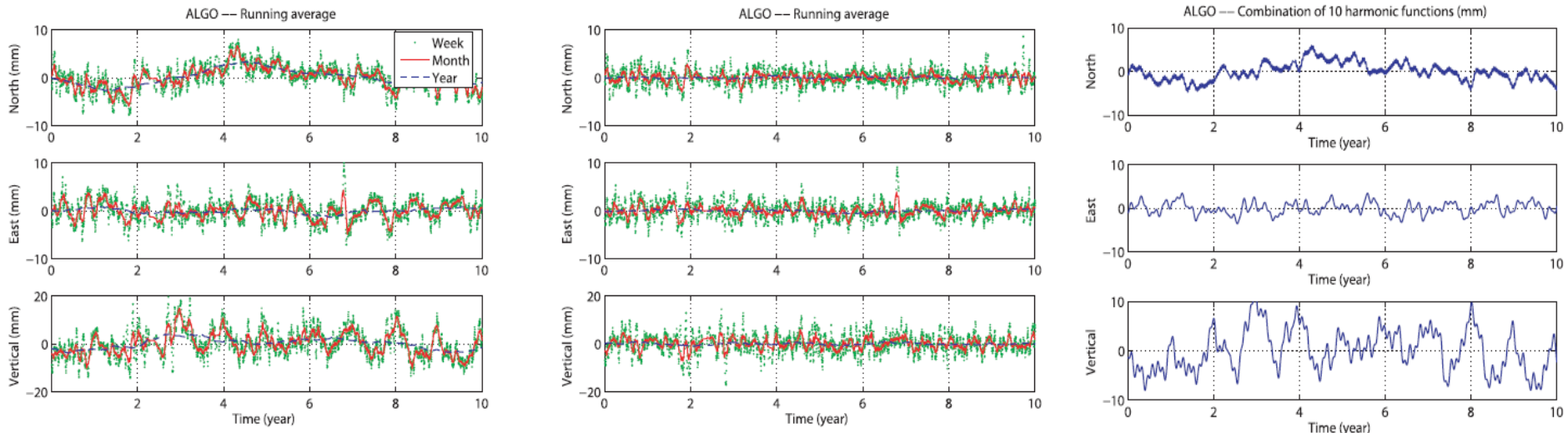


Application to GPS Position Time Series

ALGO IGS Station

Weekly, monthly, and yearly mean residual series of north, east, and vertical components for two functional models (site ALGO); linear trend only (left); and linear trend plus 10 harmonic functions (right).

Combination of 10 harmonic functions (unmodelled effects) as removed from original coordinate component time series



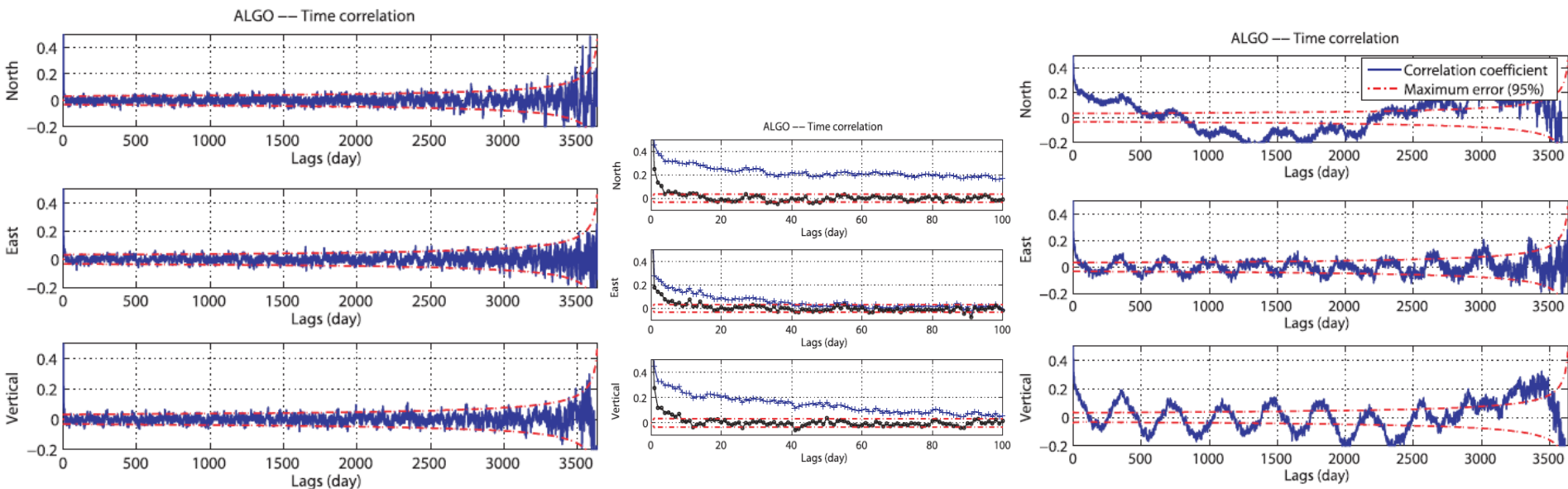
Application to GPS Position Time Series

ALGO IGS Station

Autocorrelation coefficients for time series of north, east, and vertical components before (right) and after (left) removing 10 harmonic functions.

Graphs in middle provides a “zoom in” over the first 100 days for both before (+) and after (o) removing 10 harmonic functions.

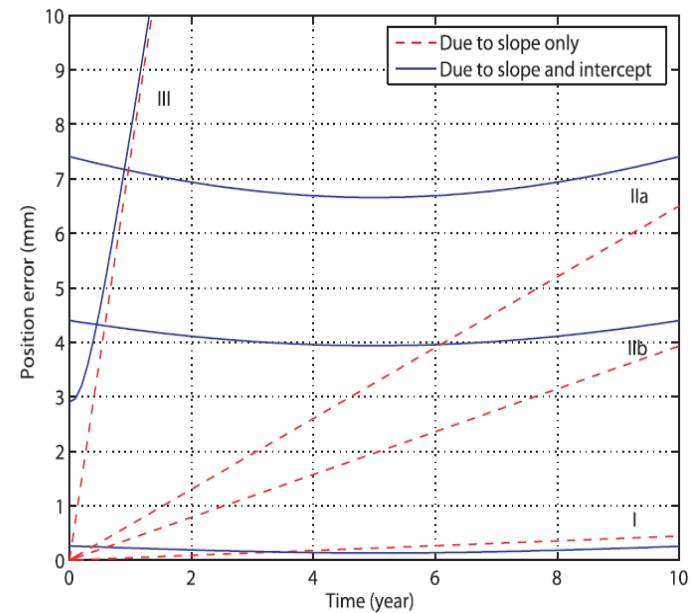
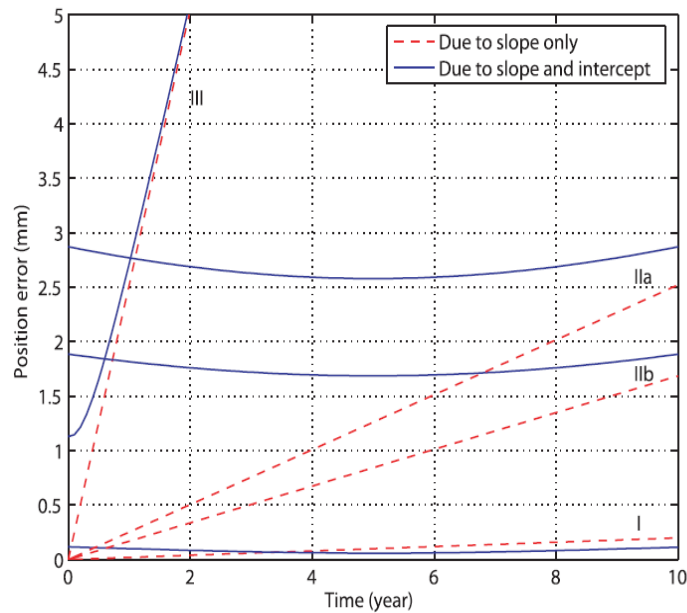
A white-like noise is observed.



Application to GPS Position Time Series

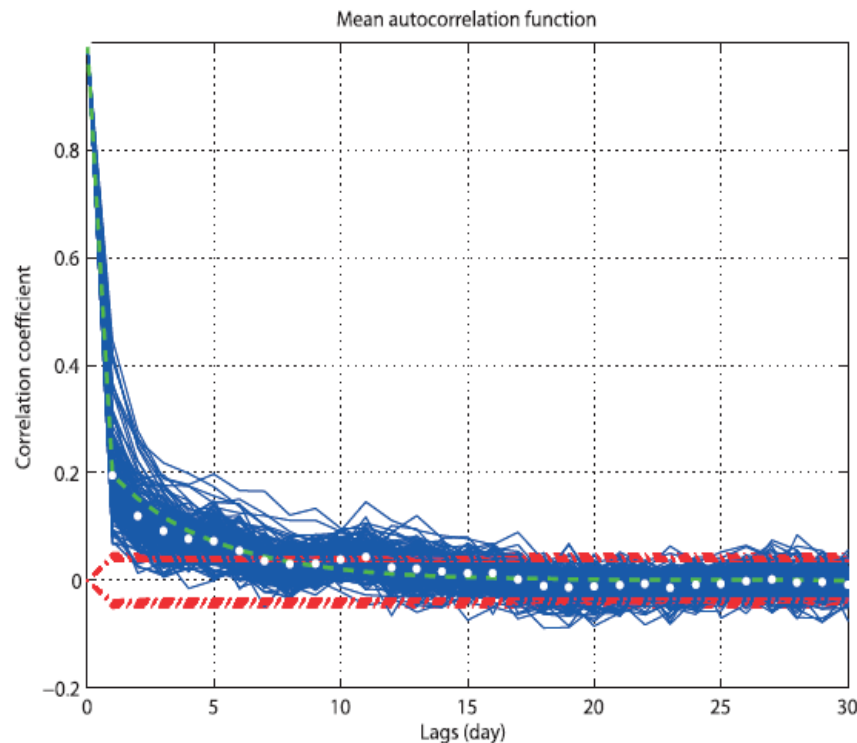
ALGO IGS Station

Position error (standard deviation) of different stochastic models as a function of time for north and vertical component of KOSG caused by rate uncertainty only (dashed lines) and full covariance matrix of intercept and site velocity (solid lines); WN only (I), WN + FN (IIa), SM (short-memory) + FN (IIb), and WN + RW (III).



Application to GPS Position Time Series

Individual autocorrelation functions (solid line) as well as mean autocorrelation function (white circles) of 71 permanent GPS stations (213 time series) for a linear regression model with 10 harmonic functions; light dashed line represents the estimated short-memory (SM = WN +AR(1)) noise process.



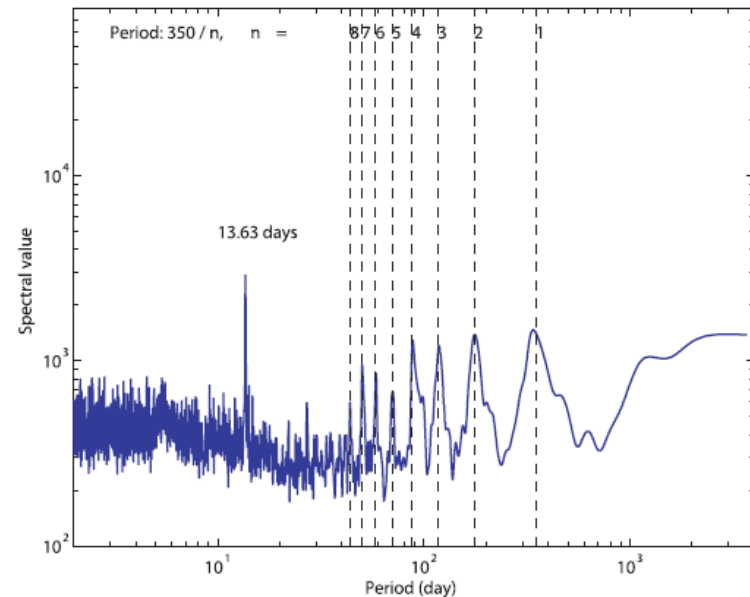
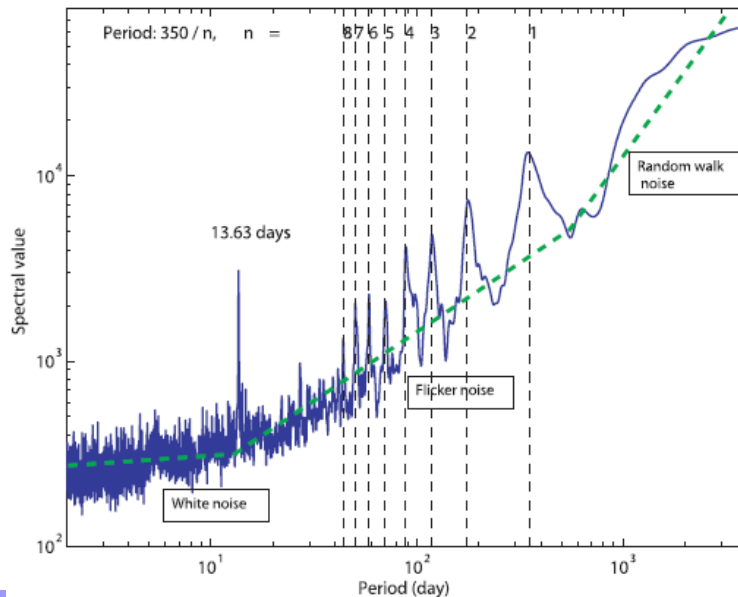
Application to GPS Position Time Series

Stacked power spectra

Stacked (weighted) power spectra of 71 permanent GPS stations (213 time series) for a linear regression model when annual and semiannual signals were removed; use of estimated pure white noise model (left); use of estimated short-memory plus flicker noise

The word “weighted” also implies that the individual spectra have been scaled according to their estimated noise magnitudes (components). Simplified form of the following equations.

$$T_{2r} = \text{tr} \left(\hat{E}^T Q^{-1} A_k (A_k^T Q^{-1} P_A^{-1} A_k)^{-1} A_k^T Q^{-1} \hat{E} \hat{\Sigma}^{-1} \right) \sim \chi^2(2r) \rightarrow P(\omega_j) = \sum_{i=1}^r \hat{e}_i^T A_j (A_j^T A_j)^{-1} A_j^T \hat{e}_i / \sigma_{ii},$$

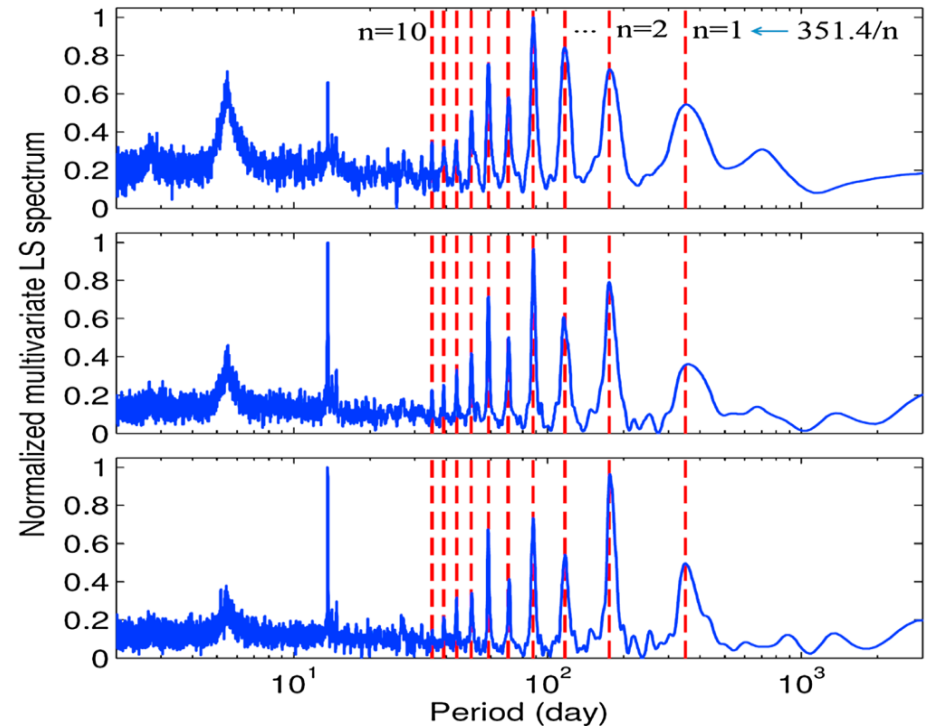


Application to GPS Position Time Series

Multivariate LS-HE

Multivariate least squares power. Vertical axes are normalized, such that the maximum power is equal to one; (top) 350 stations, (middle) 150 stations, (bottom) and 50 stations.

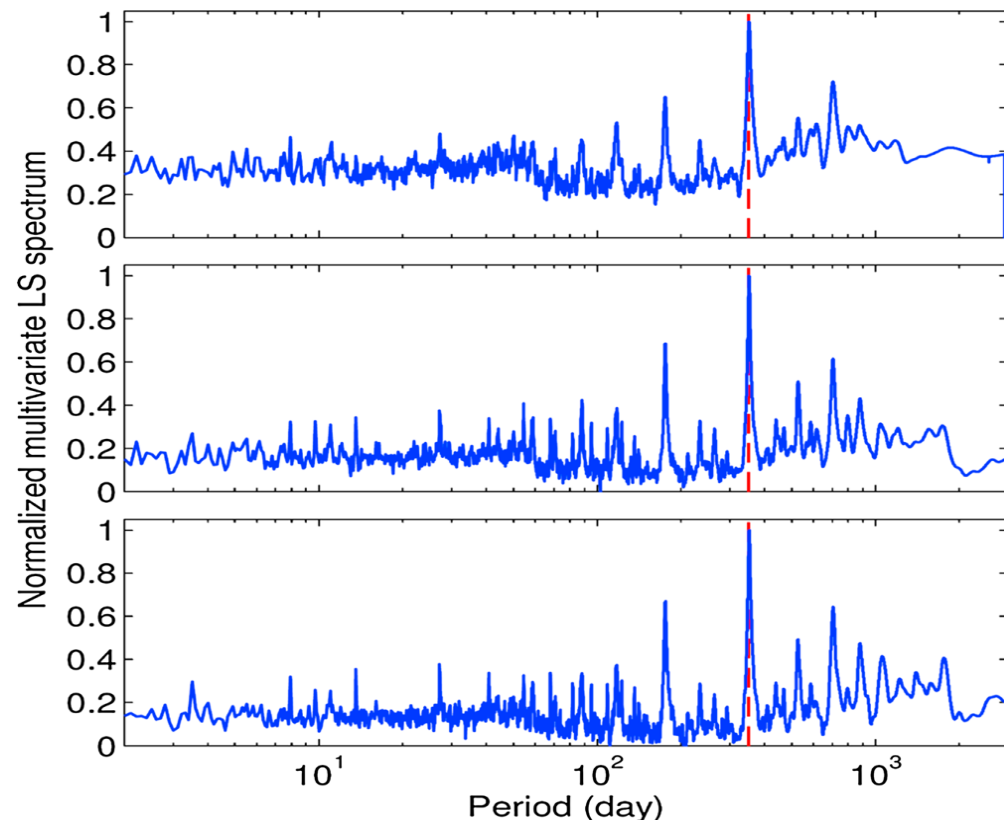
- Draconitic signals
- 13.63 days signal
- 5-6 days quasi-signals



Application to GPS Position Time Series

Multivariate multiharmonic LS-HE

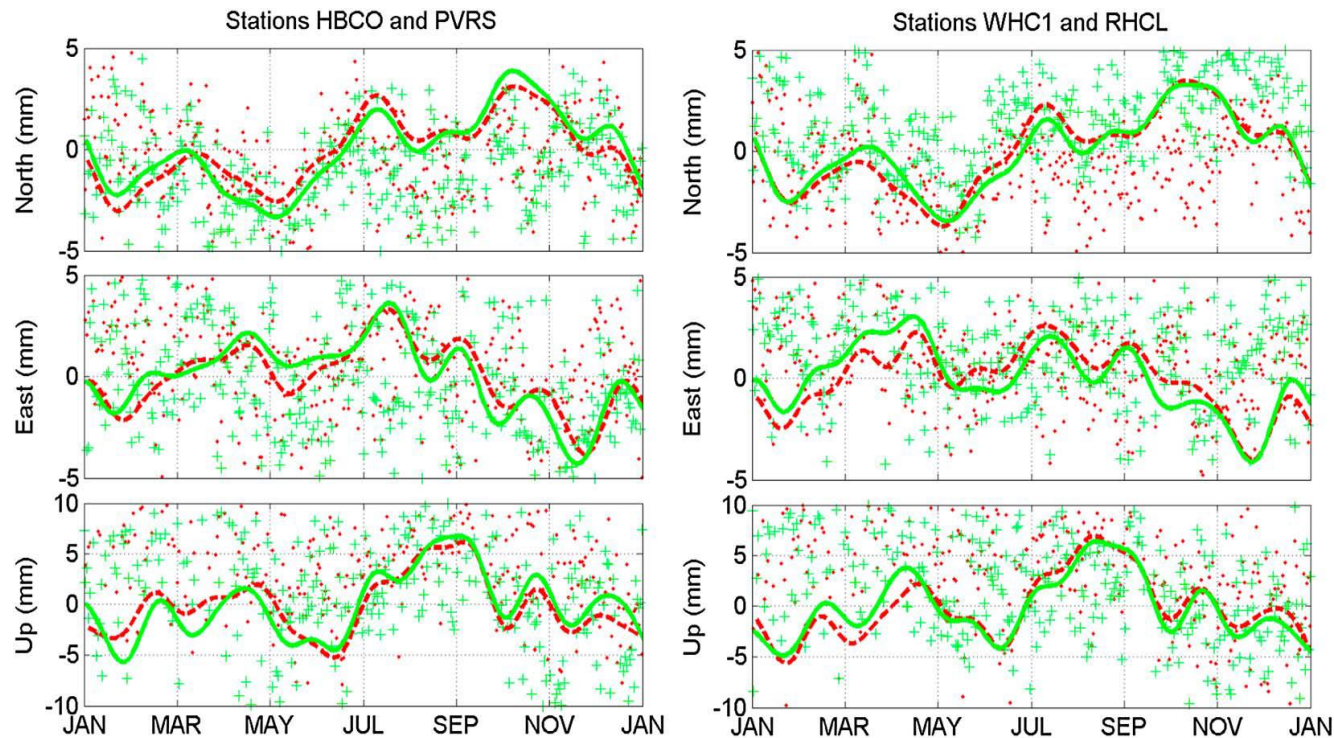
- **Multivariate multiharmonic spectrum using 10 harmonics.**
- **Vertical axes are normalized such that the maximum power is equal to one.**
- **(top) 350 stations, (middle) 150 stations, and (bottom) 50 stations.**
- **Draconitic signal is simply estimated to be 351.6 ± 2 days**



Application to GPS Position Time Series

Nature of draconitic effect

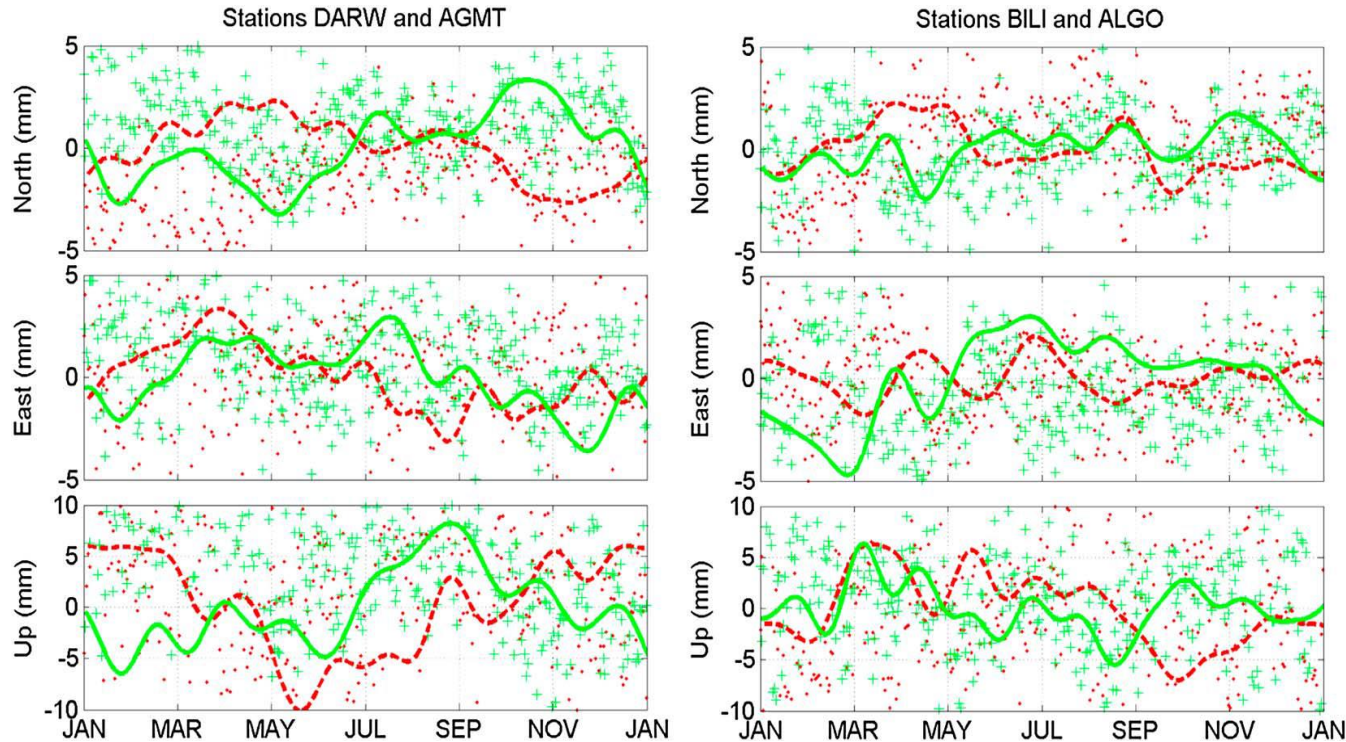
Effect of periodic pattern estimated for two typical examples (PVRs versus HBCO and RHCL versus WHC1) in which stations are close to each other.



Application to GPS Position Time Series

Nature of draconitic effect

Effect of periodic pattern estimated for two typical examples (DARW versus AGMT and BILL versus ALGO) in which stations are far from each other.

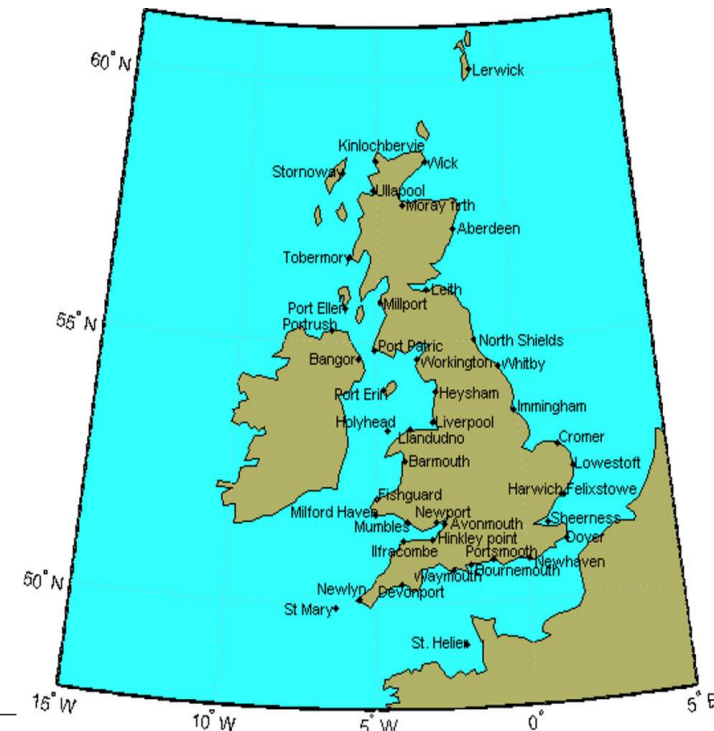


Application to tidal time series

UK tide-gauge stations

Appendix A: UK tide-gauge stations' names and coordinates

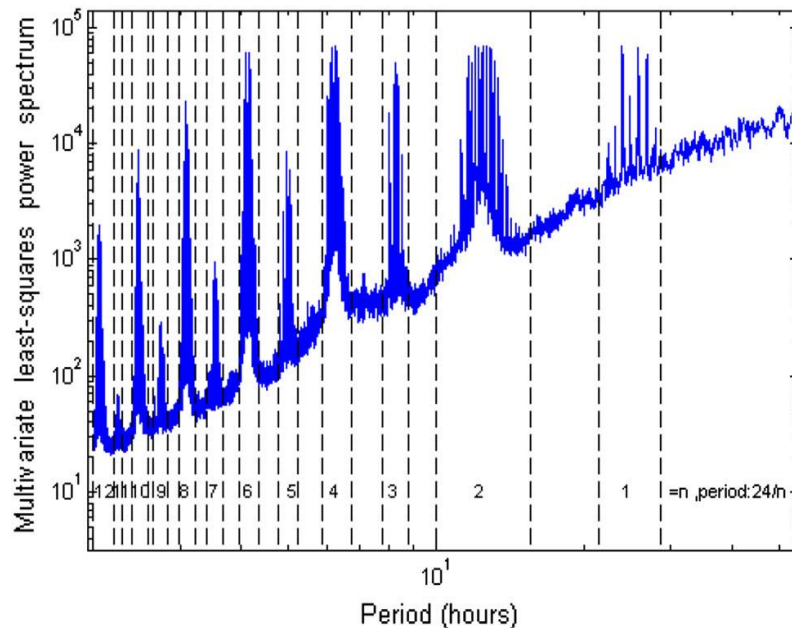
No.	St. name	Latitude (°)	Longitude (°)	No.	St. name	Latitude (°)	Longitude (°)
1	Aberdeen	57.1440	-2.0803	24	Milford Haven	51.7064	-5.0514
2	Avonmouth	51.5108	-2.7151	25	Millport	55.7496	-4.9058
3	Bangor	54.6648	-5.6695	26	Moray firth	57.5992	-4.0002
4	Barmouth	52.7193	-4.0450	27	Mumbles	51.5703	-3.9749
5	Bournemouth	50.7143	-1.8749	28	Newhaven	50.7818	0.0570
6	Cromer	52.9343	1.3016	29	Newlyn	50.1030	-5.5428
7	Devonport	50.3684	-4.1853	30	Newport	51.5500	-2.9874
8	Dover	51.1144	1.3225	31	North Shields	55.0074	-1.4398
9	Felixstowe	51.9577	1.3466	32	Port Ellen	55.6276	-6.1899
10	Fishguard	52.0137	-4.9832	33	Port Erin	54.0852	-4.7681
11	Harwich	51.9480	1.2921	34	Port Patric	54.8426	-5.1200
12	Heysham	54.0318	-2.9203	35	Portrush	55.2068	-6.6568
13	Hinkley point	51.2153	-3.1345	36	Portsmouth	50.8026	-1.1118
14	Holyhead	53.3139	-4.6206	37	St Mary	49.9185	-6.3165
15	Ilfracombe	51.2109	-4.1111	38	Sheerness	51.4456	0.7434
16	Immingham	53.6310	-0.1868	39	Stornoway	58.2070	-6.3889
17	St. Helier	49.1833	-2.1167	40	Tobermory	56.6229	-6.0640
18	Kinlochbervie	58.4567	-5.0504	41	Ullapool	57.8953	-5.1581
19	Leith	55.9898	-3.1817	42	Waymouth	50.6085	-2.4479
20	Lerwick	60.154	-1.1403	43	Whitby	54.4894	-0.6199
21	Liverpool	53.4497	-3.0181	44	Wick	58.4410	-3.0865
22	Llandudno	53.3317	-3.8252	45	Workington	54.6508	-3.5678
23	Lowestoft	52.4820	1.7516				



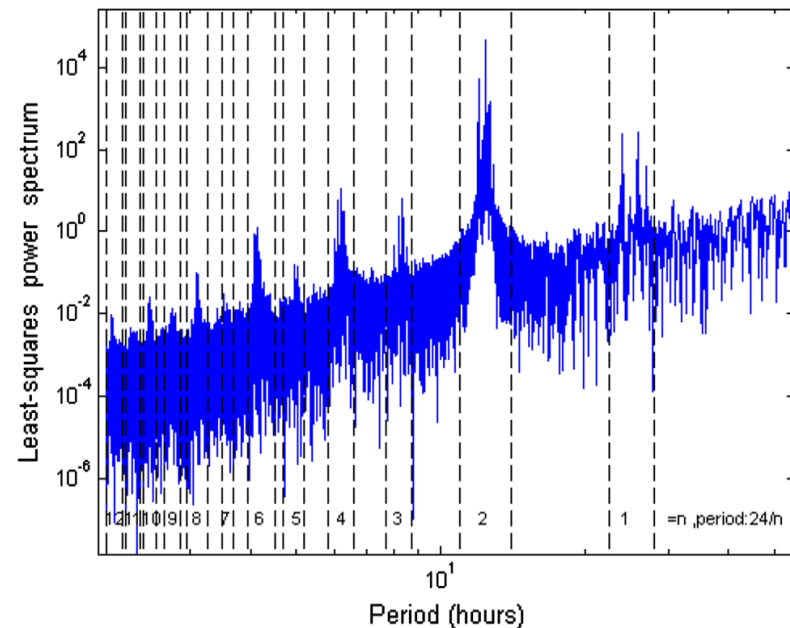
Application to tidal time series

Multivariate vs Univariate LS-HE

- **Multivariate least-squares power spectrum**
- **318 1-year tide height time series**
- **Sample rate of 15 min provided from 43 tidegauge stations in UK.**
- **The *dashed lines* indicate period range of diurnal signal and its higher harmonics**



- **Univariate least-squares power spectrum**
- **Tide height time series of station at North Shields in UK**
- **Sampling rate of 15 min**
- **The *dashed lines* indicate period range of diurnal signal and its higher harmonics**



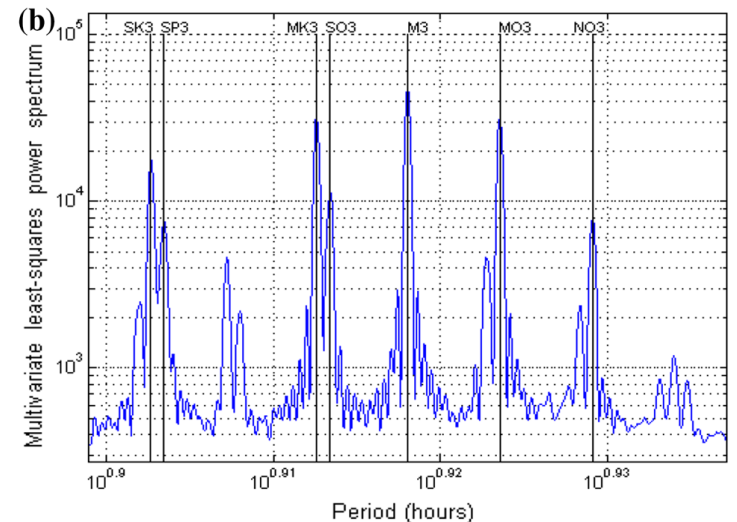
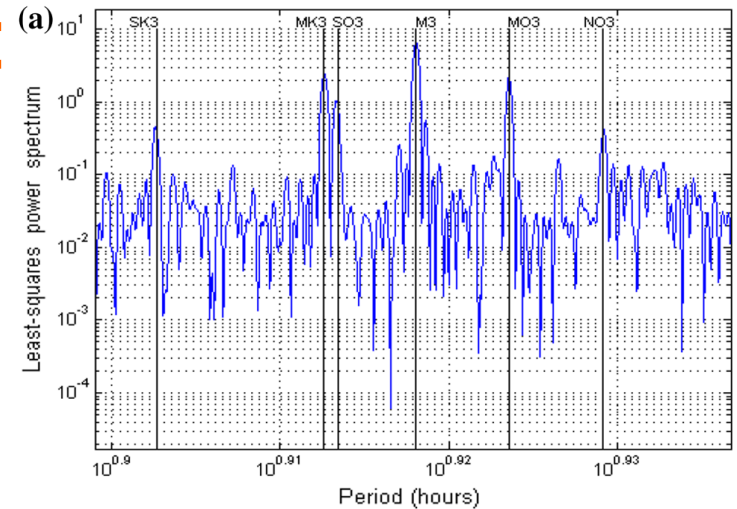
Application to tidal time series

Multivariate vs Univariate LS-HE

Zoom-in of ter-diurnal signal in
least-squares power spectrum of

(a) a single 1-year time series

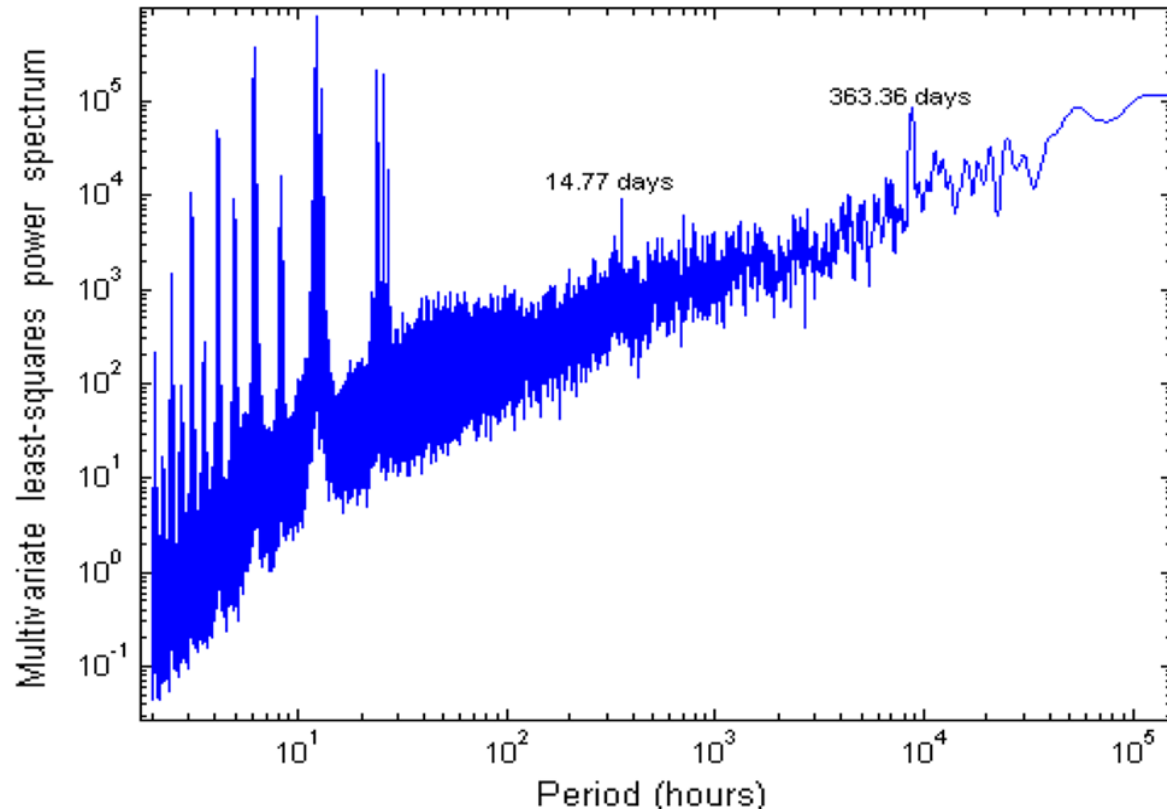
(b) 318 1-year time series



Application to tidal time series

Multivariate LS-HE

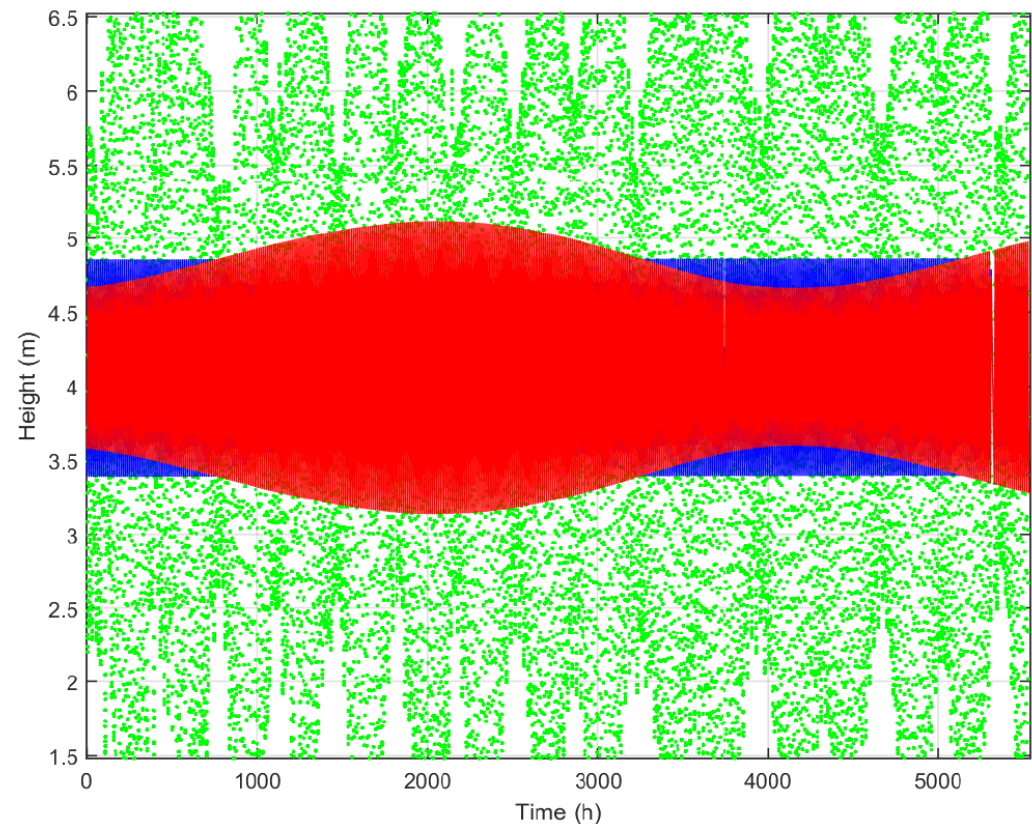
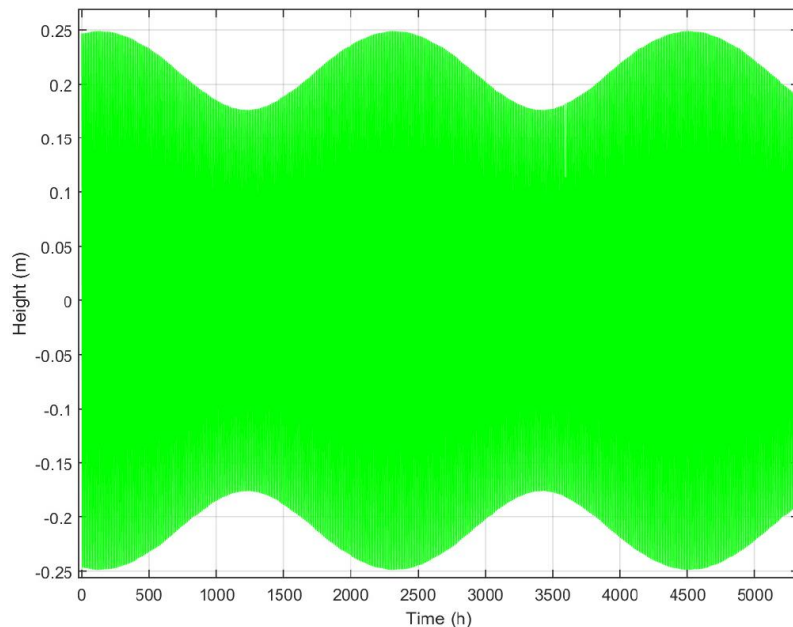
- **Multivariate least-squares power spectrum**
- **Eleven 18.24-year tide height time series**
- **Sampling rate of 15 min**
- **Provided from 11 tide-gauge stations in UK**



Application to tidal time series

Multivariate modulated signals

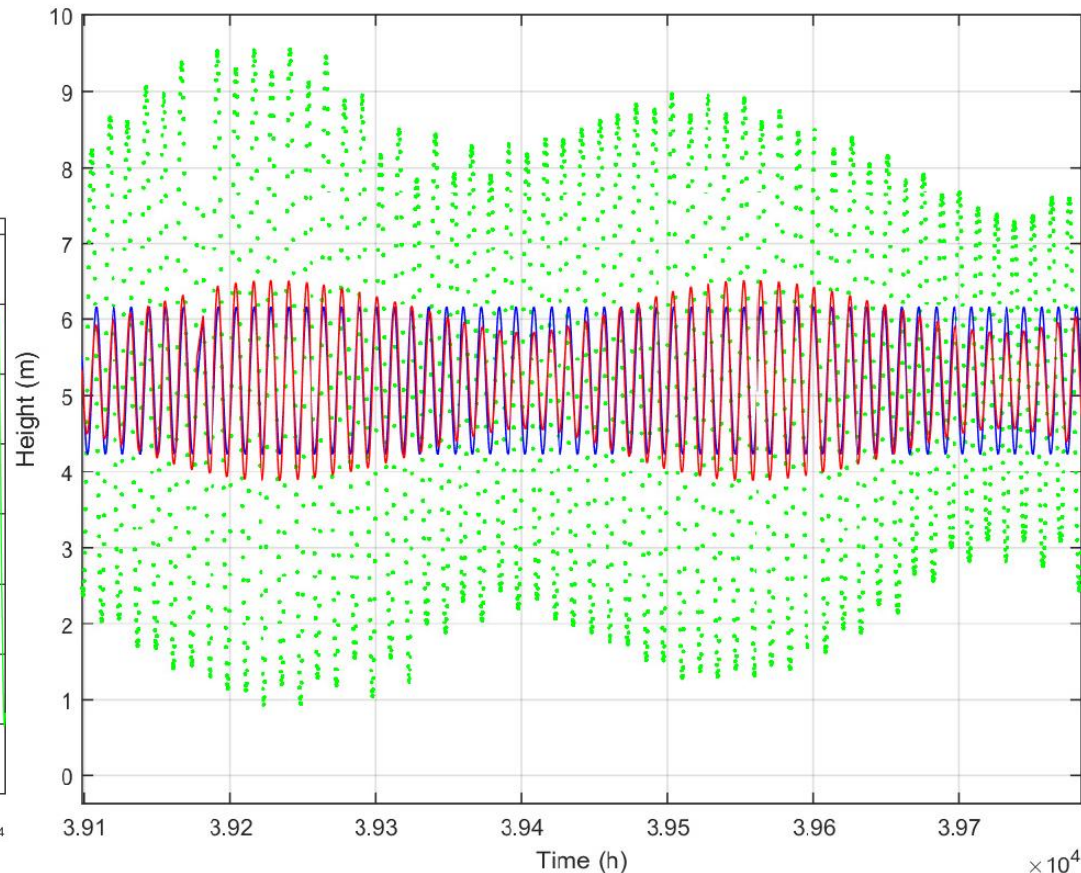
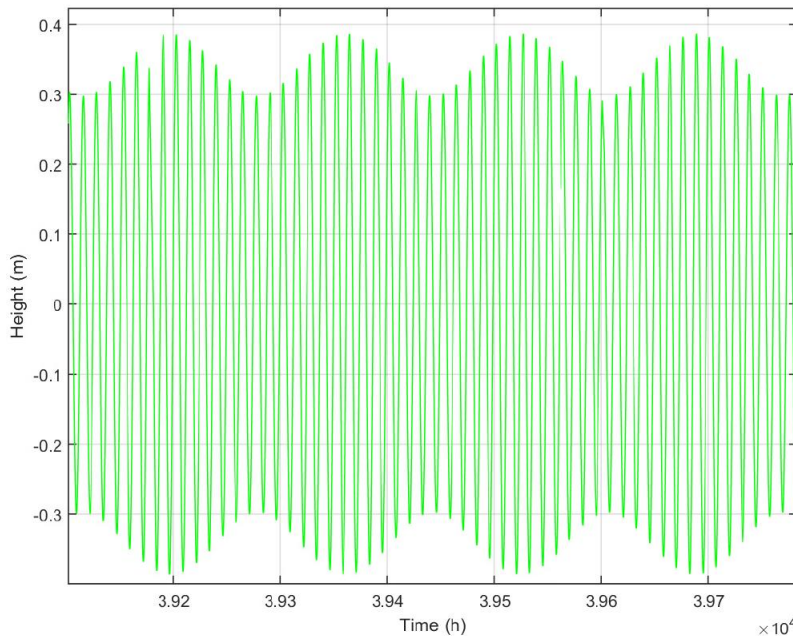
- **S2 modulated with semi-annual signal**
- **S2 signal (blue)**
- **S2 plus modulated S2 (red)**
- **Modulated S2 (green)**



Application to tidal time series

Multivariate modulated signals

- **S2 modulated with MF (13.66 days)**
- **S2 signal (blue)**
- **S2 plus modulated S2 (red)**
- **Modulated S2 (green)**



Application to tidal time series

List of tidal frequencies: MLS-HE

No.	Frequency (cycle/h)	Mean amplitude (m)	Darwin's symbol	No.	Frequency (cycle/h)	Mean amplitude (m)	Darwin's symbol
1	0.0805114	$2.122 \pm 2e-004$	M2	26	0.1207668	$0.018 \pm 2e-004$	M3
2	0.0833331	$0.735 \pm 2e-004$	S2	27	0.0802832	$0.017 \pm 2e-004$	OP2
3	0.0789996	$0.405 \pm 2e-004$	N2	28	0.0012733	$0.017 \pm 5e-004$	MSM
4	0.0835607	$0.207 \pm 2e-004$	K2	29	0.0000186	$0.017 \pm 3e-004$	—
5	0.0387306	$0.108 \pm 3e-004$	O1	30	0.0863831	$0.016 \pm 2e-004$	SKM2
6	0.0417806	$0.103 \pm 2e-004$	K1	31	0.0001511	$0.016 \pm 4e-004$	SSA
7	0.1610228	$0.100 \pm 2e-004$	M4	32	0.0000400	$0.016 \pm 4e-004$	—
8	0.0820239	$0.100 \pm 2e-004$	L2	33	0.0748673	$0.016 \pm 2e-004$	ST37
9	0.0776897	$0.087 \pm 3e-004$	MU2	34	0.0807396	$0.015 \pm 3e-004$	MKS2
10	0.0792009	$0.083 \pm 2e-004$	NU2	35	0.0028216	$0.014 \pm 2e-004$	MSF
11	0.0001150	$0.071 \pm 5e-004$	SA	36	0.0806252	$0.013 \pm 2e-004$	H2
12	0.1638446	$0.064 \pm 2e-004$	MS4	37	0.1222924	$0.013 \pm 1e-004$	MK3
13	0.0818213	$0.045 \pm 2e-004$	LDA2	38	0.1582011	$0.013 \pm 2e-004$	3MS4
14	0.1595109	$0.039 \pm 2e-004$	MN4	39	0.0030503	$0.012 \pm 4e-004$	MF
15	0.0372188	$0.035 \pm 3e-004$	Q1	40	0.1625353	$0.011 \pm 1e-004$	KN4
16	0.0415524	$0.034 \pm 2e-004$	P1	41	0.1192424	$0.011 \pm 2e-004$	MO3
17	0.0861555	$0.033 \pm 2e-004$	2SM2	42	0.0763798	$0.010 \pm 3e-004$	ST2
18	0.0848456	$0.025 \pm 2e-004$	MSN2	43	0.0000889	$0.010 \pm 4e-004$	—
19	0.0774621	$0.024 \pm 2e-004$	O2	44	0.0000480	$0.010 \pm 2e-004$	—
20	0.0014283	$0.022 \pm 5e-004$	MM	45	0.1666663	$0.010 \pm 2e-004$	S4
21	0.0761772	$0.021 \pm 3e-004$	EPS2	46	0.1623327	$0.010 \pm 2e-004$	SN4
22	0.2443561	$0.021 \pm 1e-004$	2MS6	47	0.2400219	$0.010 \pm 1e-004$	2MN6
23	0.0803976	$0.021 \pm 1e-004$	H1	48	0.1597129	$0.010 \pm 1e-004$	ST9
24	0.2415344	$0.018 \pm 1e-004$	M6	49	0.0402562	$0.009 \pm 2e-004$	NO1
25	0.1640721	$0.018 \pm 2e-004$	MK4	50	0.0759496	$0.009 \pm 2e-004$	OQ2

OP2=S2-MF

14.77 days

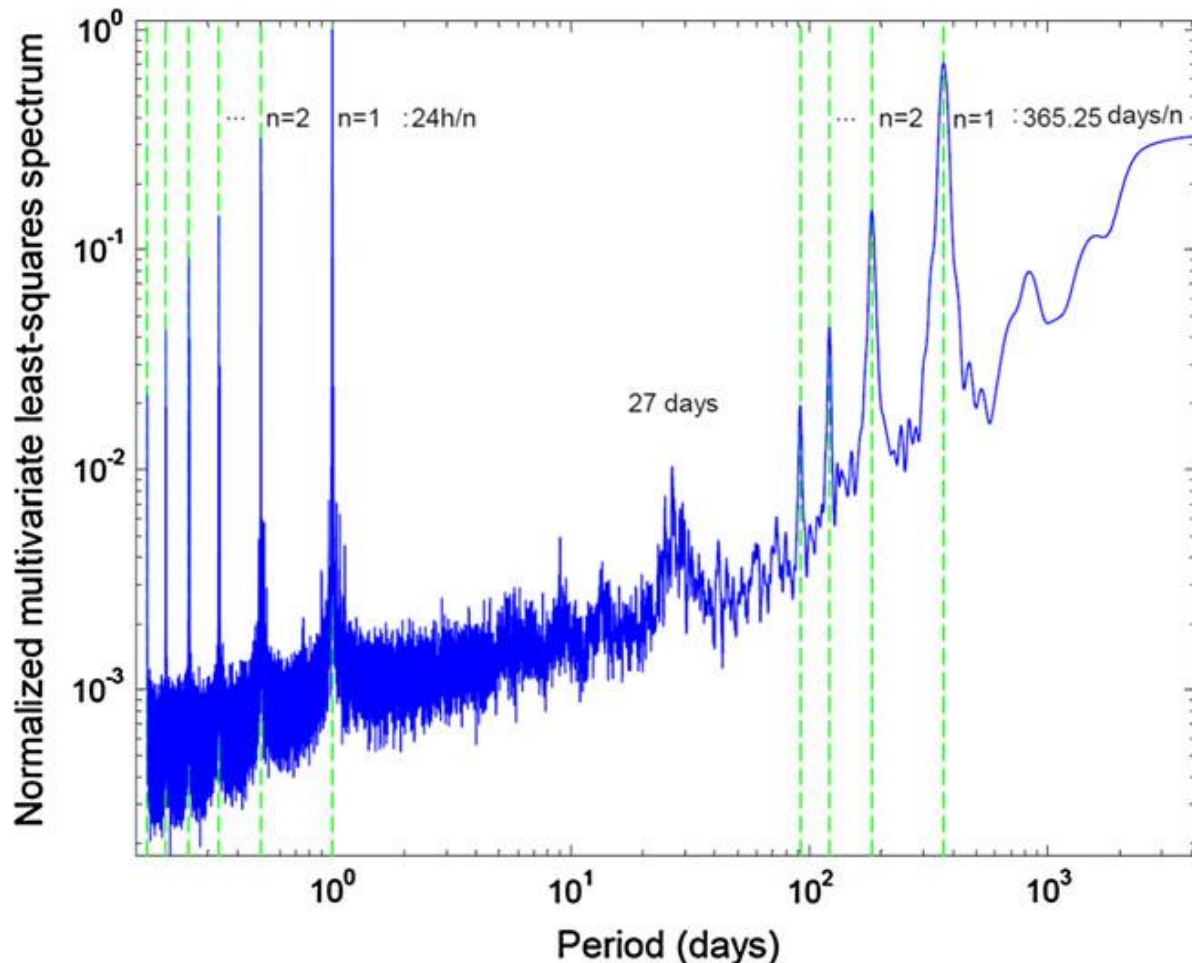
13.66 days



Application to TEC time series

Multivariate modulated signals

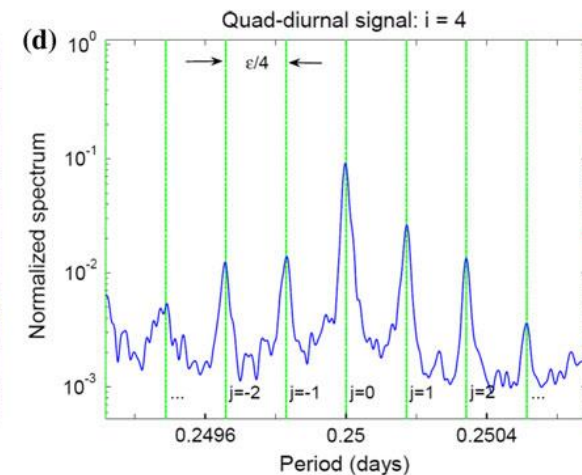
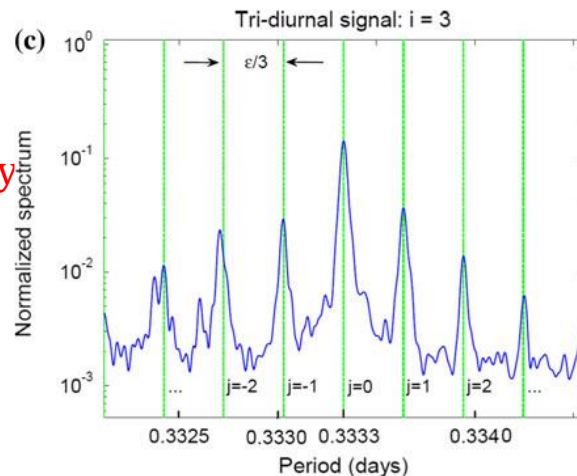
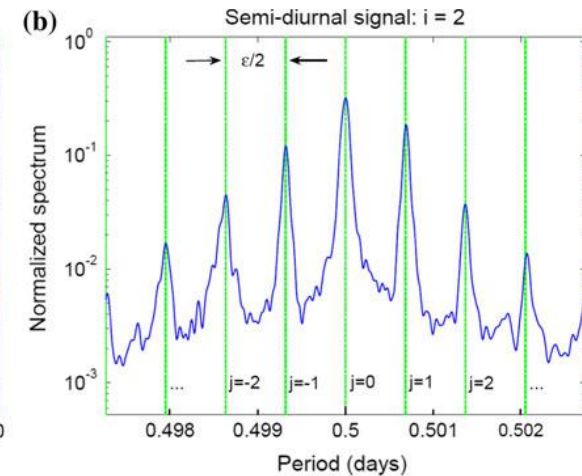
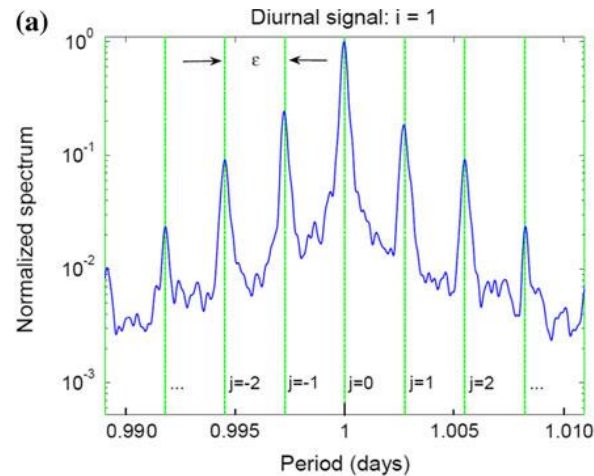
- **Multivariate power spectrum**
- **33 bihourly TEC time series**
- **Lat. ranging from -80 to 80**
- **Period 2001-2007**
- **Diurnal and its harmonics**
- **Annual and its harmonics**



Application to TEC time series

Multivariate modulated signals

- **Zoom-in of power spectrum**
- **Frequency of daily signal is**
 $\omega_i = 2\pi i$ (cycle per day)
- **Frequency of annual signals is**
 $\omega_j = \frac{2\pi j}{365.25}$ (cycle per day)



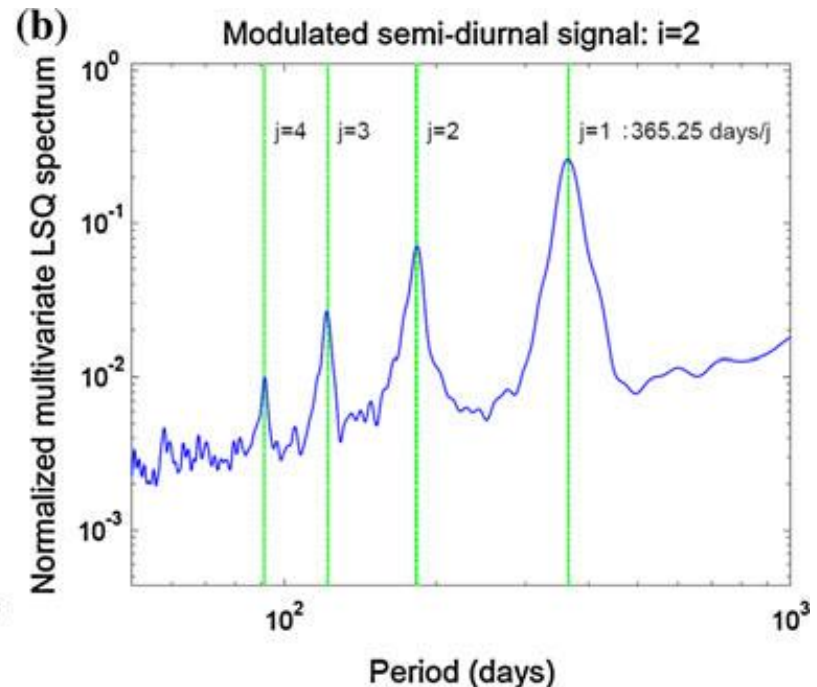
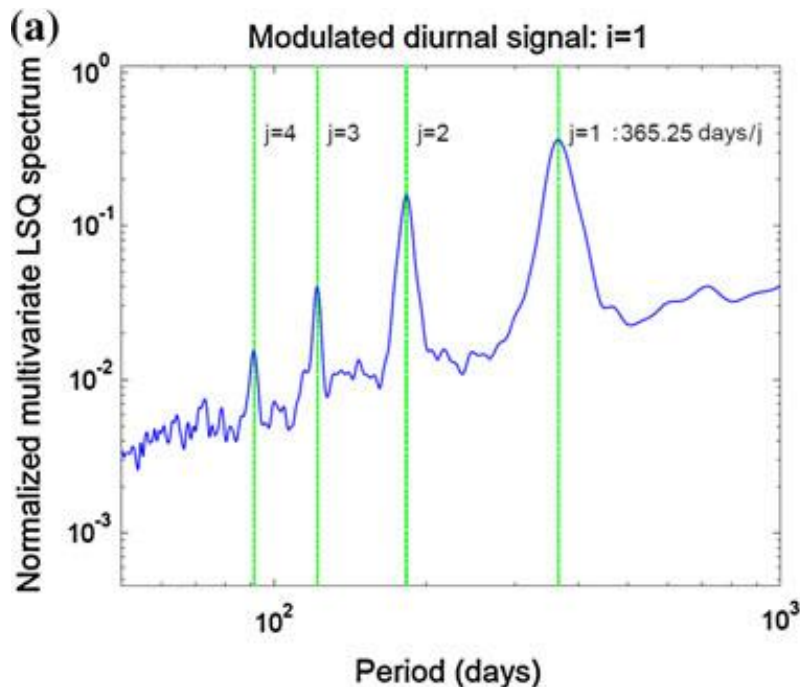
- **Modulated signals is**
 $\omega_{ij}^M = 2\pi i(1 \pm \frac{j}{365.25})$ (cycle per day)
- i and $j = 1, 2, 3, 4$



Application to TEC time series

Multivariate modulated signals

- Multivariate least-squares spectrum
- Modulated diurnal (a) and semidiurnal (b) signals
- annual ($j = 1$), semiannual ($j = 2$), triannual ($j = 3$), and quad-annual ($j = 4$) signals



Application to TEC time series

Multivariate modulated signals

- Diurnal/semidiurnal signal modulated with annual (a/b) and semiannual (c/d) signals for 2001
- Superposition of diurnal and semidiurnal signals modulated with annual/ semiannual signals ($a + b / c + d$)
- Superposition of diurnal/ semidiurnal signal modulated with annual and semiannual signals ($a + c / b + d$)
- Superposition of all four signals (frames: $a+b+c+d$)
- Vertical lines indicate equinoxes and solstices

